

LumièreShop: Enterprise Data Warehouse Architecture & Schema Reference

Dataset: ecommerce_dw

GCP Project: <YOUR_GCP_PROJECT_ID>

Location: <YOUR_GCP_REGION>

Scale: 140 Tables | 19,423,487 Total Rows | 1,765.35 MB (1.72 GB) Storage

1. Executive Data Warehouse Summary

The LumièreShop BigQuery Data Warehouse (`ecommerce_dw`) models an omnichannel European luxury beauty and retail enterprise. The dataset is partitioned into:

- 1. **Core Diagnostic Cluster (25 Tables):** High-resolution transactional and telemetry tables powering the Black Week 2026 revenue incident root-cause discovery.
- 2. **Enterprise Extended Domains (115 Tables):** Full-scope operational, accounting, logistics, marketing, machine learning, and security systems (Domains A through Q).

Storage & Volume Breakdown by Table (All 140 Tables)

Table Name	Record Count	Data Size	Primary Key	Operational Role & Domain
web_events	17,290,297	1590.80 MB	session_id	[PURPOSE]: User interaction clickstream funnel events capturing product page views, cart additions, checkout initiations, and transaction successes for conversion funnel drop-off analysis
web_sessions	1,716,000	138.00 MB	session_id	[PURPOSE]: Web clickstream traffic sessions recording marketing channels, UTM acquisition tags, device operating systems, browsers, and user journey attribution
order_items	83,788	7.06 MB	ord_hdr_num, order_item_id	[PURPOSE]: Discrete purchase line items recording product SKU references, realized sale prices, promotional discounts, and basket-level revenue realization
sales_event_stream	83,788	7.51 MB	event_id	[PURPOSE]: Real-time streaming transactional event feed capturing high-frequency sales events and intra-hour intake velocity
orders	52,365	3.35 MB	order_id	[PURPOSE]: Core transactional sales order headers capturing customer checkout completion, gross merchandise value (GMV), order status, payment confirmation, and timestamps for commercial revenue analysis
payment_gateway_logs	51,182	5.13 MB	session_id	[PURPOSE]: Payment service provider (PSP) authorization logs capturing transaction success rates, latency in milliseconds, error codes, and checkout drop-offs across Stripe, PayPal, and Adyen
inventory_snapshots	28,200	908.8 KB	product_id, distribution_center_id, snapshot_hour	[PURPOSE]: Daily and hourly historical inventory snapshots tracking stock depletion, stockouts, and inventory availability across warehouse hubs
users	10,000	964.5 KB	user_id	[PURPOSE]: Customer user accounts with demographic localization, country mapping, and lifetime activity profiles
email_send_queue_logs	5,000	197.7 KB	*(Surrogate)*	[PURPOSE]: Outbound email dispatch queue execution logs, deliverability statuses, and send timestamps
stg_ga4_clickstream_raw	5,000	352.8 KB	*(Surrogate)*	[PURPOSE]: Raw Google Analytics 4 export event records from BigQuery streaming export
stg_klaviyo_email_events_raw	5,000	727.1 KB	event_id	[PURPOSE]: Raw event stream from Klaviyo marketing automation webhook (opens, clicks, bounces)
ticket_messages	5,000	436.0 KB	*(Surrogate)*	[PURPOSE]: Detailed communication thread messages exchanged between support agents and customers
live_chat_messages	4,000	437.6 KB	*(Surrogate)*	[PURPOSE]: Individual real-time chat transcript messages exchanged during live customer chat sessions
loyalty_points_ledger	4,000	222.7 KB	*(Surrogate)*	[PURPOSE]: Detailed transactional audit ledger recording loyalty points earned

Table Name	Record Count	Data Size	Primary Key	Operational Role & Domain
				from purchases, bonus campaigns, and point expirations
stg_stripe_payment_intents_raw	4,000	306.7 KB	*(Surrogate)*	[PURPOSE]: Raw Stripe payment intent webhook JSON payloads for credit card authorizations
catalog_recommender_logs	3,250	248.8 KB	session_id	[PURPOSE]: On-page product recommendation widget impressions capturing algorithm fallback events, category mismatch errors, and lost substitution sales
category_15min_targets	3,072	171.0 KB	category_id, interval_start_time	[PURPOSE]: Intraday 15-minute pacing targets modeling hourly customer traffic waves, intake velocity benchmarks, and intra-day pacing curves
loyalty_members	2,500	97.7 KB	*(Surrogate)*	[PURPOSE]: Customer loyalty program membership profiles, reward tier standings (Silver, Gold, Platinum), and active point balances
pos_transaction_items	2,500	97.7 KB	*(Surrogate)*	[PURPOSE]: Granular purchase line items sold through physical store cash registers
push_notification_receipts	2,500	97.6 KB	*(Surrogate)*	[PURPOSE]: Individual mobile app push notification delivery receipts and user tap/click interactions
stg_ga4_traffic_sources_raw	2,500	138.4 KB	session_id	[PURPOSE]: Raw GA4 traffic acquisition attribution source records
stg_shopify_orders_raw	2,500	368.0 KB	*(Surrogate)*	[PURPOSE]: Raw Shopify webhook JSON payloads capturing e-commerce checkout and order events
stg_zendesk_tickets_raw	2,500	171.7 KB	*(Surrogate)*	[PURPOSE]: Raw customer support ticket JSON records extracted from Zendesk Support API
user_subscription_preferences	2,500	65.9 KB	*(Surrogate)*	[PURPOSE]: Customer communication consent settings, marketing opt-ins (email, SMS, push), and GDPR compliance timestamps
product_attribute_values	2,000	72.6 KB	*(Surrogate)*	[PURPOSE]: EAV (Entity-Attribute-Value) structured technical product specifications and filter criteria for catalog search
sms_delivery_receipts	2,000	84.0 KB	*(Surrogate)*	[PURPOSE]: Telecommunications carrier SMS delivery receipts and handset delivery confirmations
stg_wms_shipments_raw	2,000	169.9 KB	*(Surrogate)*	[PURPOSE]: Raw warehouse management system (WMS) carrier dispatch manifests
store_inventory_levels	2,000	78.1 KB	*(Surrogate)*	[PURPOSE]: Real-time store on-hand stock quantities, shelf availability, and periodic cycle count audit dates per retail location
support_tickets	2,000	135.3 KB	*(Surrogate)*	[PURPOSE]: Customer service support cases, ticket priority, omnichannel resolution duration, and first-response metrics

Table Name	Record Count	Data Size	Primary Key	Operational Role & Domain
coupon_redemption_audit	1,500	70.3 KB	*(Surrogate)*	[PURPOSE]: Audit logs recording every checkout coupon code application, user ID, and monetary discount amount realized in EUR
csat_surveys	1,500	85.0 KB	*(Surrogate)*	[PURPOSE]: Post-service customer satisfaction survey scores (1-5 scale) and verbatim feedback submitted following ticket resolution
customer_refunds	1,500	87.4 KB	*(Surrogate)*	[PURPOSE]: Monetary refund disbursements issued to customers following verified returns, cancellations, or damaged item claims
dev_customer_churn_feature_store	1,500	87.9 KB	*(Surrogate)*	[PURPOSE]: Machine learning feature store containing customer behavioral aggregations and predicted churn risk scores
legacy_orders_2023_archive	1,500	48.4 KB	*(Surrogate)*	[PURPOSE]: Deprecated archive of historical 2023 customer order headers retained for legal and compliance audit
live_chat_sessions	1,500	117.2 KB	*(Surrogate)*	[PURPOSE]: Real-time live chat sessions between web visitors and support agents, recording queue wait times and chat duration
pallet_inventory_locations	1,500	71.8 KB	*(Surrogate)*	[PURPOSE]: Real-time pallet License Plate Number (LPN) tracking linking physical pallets to warehouse bin coordinates
pos_store_transactions	1,500	91.2 KB	*(Surrogate)*	[PURPOSE]: In-store brick-and-mortar point-of-sale customer sales transactions, cashier employee IDs, and payment tender types
product_media_gallery	1,500	124.4 KB	*(Surrogate)*	[PURPOSE]: High-resolution product images, video URLs, CDN asset paths, and display sort orders
product_multilingual_translations	1,500	191.7 KB	*(Surrogate)*	[PURPOSE]: Localized product titles and marketing descriptions translated into French, German, Italian, Spanish, and Dutch
product_returns	1,500	86.6 KB	*(Surrogate)*	[PURPOSE]: Customer post-delivery return authorizations (RMA), return reasons, processing statuses, and item condition triage for reverse logistics and refund operations
return_inspections	1,500	74.7 KB	*(Surrogate)*	[PURPOSE]: Physical warehouse inspection triage logs evaluating item condition and restocking eligibility for returned goods
return_shipping_labels	1,500	98.1 KB	*(Surrogate)*	[PURPOSE]: Logistics return shipping waybills and tracking records generated for customer merchandise returns
stg_shopify_customers_raw	1,500	157.7 KB	*(Surrogate)*	[PURPOSE]: Raw Shopify customer account update webhook JSON payloads
stgZendeskSatisfactionRaw	1,500	77.0 KB	*(Surrogate)*	[PURPOSE]: Raw customer satisfaction rating survey responses from Zendesk

Table Name	Record Count	Data Size	Primary Key	Operational Role & Domain
purchase_order_line_items	1,200	56.2 KB	*(Surrogate)*	[PURPOSE]: Itemized procurement line items specifying ordered quantities, received quantities, and unit wholesale costs in EUR
qa_load_test_sessions_backup	1,200	57.5 KB	*(Surrogate)*	[PURPOSE]: Archived load test benchmark traffic logs generated by synthetic load testing tools
stg_sap_erp_inventory_feed_raw	1,200	66.9 KB	*(Surrogate)*	[PURPOSE]: Raw nightly inventory stock feed export from SAP ERP enterprise system
stg_shopify_products_raw	1,200	136.7 KB	*(Surrogate)*	[PURPOSE]: Raw Shopify product synchronization JSON payloads from catalog webhooks
oos_interactions	1,182	51.8 KB	interaction_id	[PURPOSE]: Customer browsing and cart interactions on out-of-stock items, capturing unfulfilled demand and estimated lost revenue in EUR due to inventory stockouts
agent_interaction_logs	852	4.60 MB	interaction_id	[PURPOSE]: Conversational Analytics session audit logs recording natural language prompts, generated SQL queries, latency, and token execution costs
forklift_telemetry_logs	600	23.4 KB	*(Surrogate)*	[PURPOSE]: Material handling equipment (forklift) IoT telemetry, battery state of charge, odometer readings, and maintenance alerts
inventory_items	600	28.1 KB	inventory_item_id	[PURPOSE]: Real-time master stock allocations, safety stock warning thresholds, and warehouse batch availability tracking across distribution centers
loyalty_reward_redemptions	600	27.5 KB	*(Surrogate)*	[PURPOSE]: Member reward redemption events converting accumulated loyalty points into discount vouchers or merchandise gifts
products	600	45.5 KB	product_id	[PURPOSE]: Master catalog of all retail products, default selling prices, cost of goods, brands, and category mappings for commercial intake tracking
referral_program_invites	600	33.7 KB	*(Surrogate)*	[PURPOSE]: Customer referral invitation tracking, unique referral links, and recipient invitation delivery logs
stg_google_ads_search_terms_raw	600	40.7 KB	*(Surrogate)*	[PURPOSE]: Raw Google Search Ads query term performance extracts
supplier_lead_time_history	600	28.1 KB	*(Surrogate)*	[PURPOSE]: Historical supplier manufacturing and transit lead times comparing promised days vs actual delivery dates
competitor_price_feed	500	22.9 KB	*(Surrogate)*	[PURPOSE]: Daily competitor retail pricing scrapes benchmarking price parity indices, market elasticity, and competitor price movements

Table Name	Record Count	Data Size	Primary Key	Operational Role & Domain
knowledge_base_articles	300	28.9 KB	*(Surrogate)*	[PURPOSE]: Help center self-service FAQ articles, view counts, and customer helpfulness voting metrics
legacy_products_deprecated	300	15.4 KB	*(Surrogate)*	[PURPOSE]: Discontinued catalog products and end-of-life merchandise from previous retail seasons
qa_checkout_synthetic_fuzz_tests	300	11.9 KB	*(Surrogate)*	[PURPOSE]: Automated checkout synthetic fuzz testing payloads and HTTP response code validations
replacement_orders	300	14.9 KB	*(Surrogate)*	[PURPOSE]: Zero-cost replacement orders dispatched to customers for lost, defective, or incorrect shipments
stg_sap_erp_purchase_orders_raw	300	21.5 KB	*(Surrogate)*	[PURPOSE]: Raw SAP ERP procurement purchase order dumps
stg_stripe_disputes_raw	300	18.4 KB	*(Surrogate)*	[PURPOSE]: Raw Stripe credit card chargeback disputes and inquiry logs
test_carrier_webhook_payloads	300	31.3 KB	*(Surrogate)*	[PURPOSE]: Synthetic logistics carrier webhook testing fixtures for development mocking and contract verification
test_fraud_mock_transactions	300	7.5 KB	*(Surrogate)*	[PURPOSE]: Synthetic fraud model test transactions evaluating rule triggers and mock transaction risks
gl_journal_lines	250	9.3 KB	*(Surrogate)*	[PURPOSE]: Granular debit and credit line item postings mapped to chart of accounts codes for general ledger reconciliation
accounts_payable_invoices	200	13.1 KB	*(Surrogate)*	[PURPOSE]: Vendor accounts payable (AP) commercial invoices, invoice amounts in EUR, payment due dates, and approval statuses
customs_and_duties_declarations	200	10.2 KB	*(Surrogate)*	[PURPOSE]: International import customs declarations, Harmonized System (HS) tariff codes, and paid import duty amounts in EUR
daily_category_targets	200	16.7 KB	category_id, date	[PURPOSE]: Intra-week daily pacing targets, budget allocations, expected conversion rates, and commercial revenue quotas per category for daily sales pacing and commercial performance analysis
referral_reward_claims	200	7.8 KB	*(Surrogate)*	[PURPOSE]: Referral rewards credited to advocating customers upon successful order completion by referred friends
store_employee_rosters	200	10.6 KB	*(Surrogate)*	[PURPOSE]: In-store retail employee staffing rosters, shift schedules, and operational store roles (Cashier, Floor Supervisor)
warranty_claims	200	12.5 KB	*(Surrogate)*	[PURPOSE]: Customer warranty claims, defect reports, manufacturer claim filings, and resolution triage
daily_ad_performance	173	10.8 KB	campaign_id, date	[PURPOSE]: Daily marketing performance tracking impressions, clicks, advertising spend in EUR,

Table Name	Record Count	Data Size	Primary Key	Operational Role & Domain
				attributed conversions, average CPC, and ROAS efficiency
warehouse_aisles_and_racks	160	7.0 KB	*(Surrogate)*	[PURPOSE]: Storage bin locations, aisle coordinates, vertical rack levels, and maximum weight capacity limits
accounts_payable_disbursements	150	6.3 KB	*(Surrogate)*	[PURPOSE]: Outbound cash disbursement payments executed to settle vendor accounts payable invoices
call_center_recordings_metadata	150	13.0 KB	*(Surrogate)*	[PURPOSE]: Inbound telephony call records, interactive voice response (IVR) menu traversal, call duration, and customer phone hashes
cross_dock_transfer_orders	150	8.2 KB	*(Surrogate)*	[PURPOSE]: Inter-hub inventory transfer shipments moving stock between regional distribution centers across Europe
dev_product_embedding_vectors	150	13.3 KB	*(Surrogate)*	[PURPOSE]: Product catalog semantic embedding vector embeddings for experimental visual similarity models
email_bounces_and_complaints	150	6.5 KB	*(Surrogate)*	[PURPOSE]: Hard/soft email bounce records, spam complaints, and invalid email addresses for sender reputation management
nps_feedback_responses	150	9.7 KB	*(Surrogate)*	[PURPOSE]: Relationship Net Promoter Score (NPS) surveys measuring overall customer brand loyalty and willingness to recommend
restocking_fee_logs	150	4.7 KB	*(Surrogate)*	[PURPOSE]: Restocking fee deductions applied to customer return refunds for out-of-policy returns or bulky freight items
stg_meta_ad_insights_raw	150	9.2 KB	*(Surrogate)*	[PURPOSE]: Raw Meta Marketing API adset insights JSON feeds
supplier_quality_scorecards	150	7.2 KB	*(Surrogate)*	[PURPOSE]: Monthly supplier performance evaluation scorecards tracking On-Time In-Full (OTIF %) delivery and defect rates
suppliers_master	150	7.5 KB	*(Surrogate)*	[PURPOSE]: Global product manufacturer and wholesale supplier master profiles, country codes, payment terms, and active contract status
customer_escalations	125	8.9 KB	*(Surrogate)*	[PURPOSE]: High-priority customer escalations handled by management, documenting escalation reason, resolution category, and financial goodwill concessions in EUR
general_ledger_journal_entries	125	12.6 KB	*(Surrogate)*	[PURPOSE]: Official enterprise general ledger accounting journal header entries recording double-entry bookkeeping debits and credits for monthly corporate financial closes
stg_google_ads_campaigns_raw	125	10.1 KB	*(Surrogate)*	[PURPOSE]: Raw Google Ads campaign performance reporting extracts
stg_klaviyo_campaigns_raw	125	9.0 KB	*(Surrogate)*	[PURPOSE]: Raw email campaign configuration dumps from Klaviyo REST

Table Name	Record Count	Data Size	Primary Key	Operational Role & Domain
				API
support_agents	125	5.9 KB	*(Surrogate)*	[PURPOSE]: Customer support representative profiles, tier rankings, language capabilities, and assigned shift schedules
gift_card_transactions	120	4.7 KB	*(Surrogate)*	[PURPOSE]: Individual debit and top-up transactions charged against customer gift card balances during checkout
accounts_receivable_invoices	100	5.2 KB	*(Surrogate)*	[PURPOSE]: B2B corporate client billing invoices, accounts receivable balances, credit terms, and payment settlement tracking
affiliate_commission_payouts	100	4.1 KB	*(Surrogate)*	[PURPOSE]: Monthly affiliate publisher commission calculations, attributed sales volume in EUR, and payout approval records
affiliate_publishers_directory	100	5.0 KB	*(Surrogate)*	[PURPOSE]: Third-party affiliate publisher networks, promotional partners, and contracted commission rates
stg_criteo_retargeting_raw	100	4.7 KB	*(Surrogate)*	[PURPOSE]: Raw Criteo dynamic product retargeting daily performance reporting
store_cash_drawer_counts	100	4.7 KB	*(Surrogate)*	[PURPOSE]: Daily store register cash drawer end-of-shift balancing logs and physical cash counting variances in EUR
shipping_lead_times	94	9.6 KB	*(Surrogate)*	[PURPOSE]: Fulfillment center operational metrics, carrier workload, promised delivery SLAs, and delivery delay impacts on cart abandonment and conversion
click_and_collect_orders	80	4.1 KB	*(Surrogate)*	[PURPOSE]: Buy Online Pick Up in Store (BOPIS) orders, store pickup verification PINs, customer collection timestamps, and fulfillment states
gift_card_master	80	4.9 KB	*(Surrogate)*	[PURPOSE]: Electronic and physical gift card registry, initial balances in EUR, current remaining balances, and activation states
inbound_dock_appointments	80	4.8 KB	*(Surrogate)*	[PURPOSE]: Warehouse inbound freight dock door appointment schedules, carrier arrival timestamps, and trailer unloading durations
sandbox_dynamic_pricing_sim_v1	80	3.1 KB	*(Surrogate)*	[PURPOSE]: Offline dynamic pricing simulator sandbox estimating price elasticity curves and revenue trade-offs
stg_trustpilot_reviews_raw	80	10.2 KB	*(Surrogate)*	[PURPOSE]: Raw Trustpilot public customer review feed webhooks
store_credit_issuances	80	3.8 KB	*(Surrogate)*	[PURPOSE]: Store credit ledger records, customer credit balances, and credit expiration tracking
agent_worklog_shifts	75	3.1 KB	*(Surrogate)*	[PURPOSE]: Daily customer support agent productivity records, tickets resolved, and average handle time (AHT) in seconds

Table Name	Record Count	Data Size	Primary Key	Operational Role & Domain
purchase_orders	60	3.4 KB	*(Surrogate)*	[PURPOSE]: Commercial procurement purchase order headers issued to product suppliers with contractual delivery deadlines and values in EUR
sandbox_search_ranking_ab_test	60	3.5 KB	*(Surrogate)*	[PURPOSE]: Search ranking algorithm A/B experiment evaluation logs recording NDCG@10 relevance scores
warehouse_labor_shifts	60	3.4 KB	*(Surrogate)*	[PURPOSE]: Warehouse worker labor shift schedules, picking tasks completed, and units picked per hour for logistics fulfillment
warehouse_refurbishments	60	3.0 KB	*(Surrogate)*	[PURPOSE]: Refurbishment work orders, technician labor hours, and replacement parts costs for returned open-box electronics
weekly_commercial_targets	28	1.3 KB	category_id, week_start_date	[PURPOSE]: Executive commercial financial planning benchmarks, expected visitor sessions, target revenue in EUR, and conversion rate (CVR) goals by product category and calendar week for sales target tracking and commercial performance analysis
ad_bidding_log	27	3.0 KB	log_id	[PURPOSE]: Automated ad platform bidding engine telemetry capturing target ROAS constraints, budget throttling events, and algorithmic learning phase status
currency_exchange_rates_daily	20	0.6 KB	*(Surrogate)*	[PURPOSE]: Daily foreign currency exchange rates benchmarking EUR against USD, GBP, CHF, and other major trading currencies
chart_of_accounts	10	0.4 KB	*(Surrogate)*	[PURPOSE]: Master general ledger chart of accounts defining corporate asset, liability, equity, revenue, and expense account codes
vat_period_filing_reports	7	0.3 KB	*(Surrogate)*	[PURPOSE]: Periodic statutory VAT tax return filings summarizing taxable sales revenue and collected output VAT across European jurisdictions
vat_tax_jurisdictions	7	0.2 KB	*(Surrogate)*	[PURPOSE]: Master European Value Added Tax (VAT) rate tables across European destination countries and standard/reduced rate categories
ad_creatives	5	0.4 KB	*(Surrogate)*	[PURPOSE]: Creative asset performance tracking creative fatigue, quality scores, click-through rates, and learning-limited states impacting ad delivery
bank_account_reconciliation	5	0.3 KB	*(Surrogate)*	[PURPOSE]: Monthly corporate bank account balance reconciliations comparing statement ending balances against general ledger cash
pos_terminal_registers	5	0.2 KB	*(Surrogate)*	[PURPOSE]: Point of Sale (POS) checkout terminal registers, hardware models, serial numbers, and IP address bindings

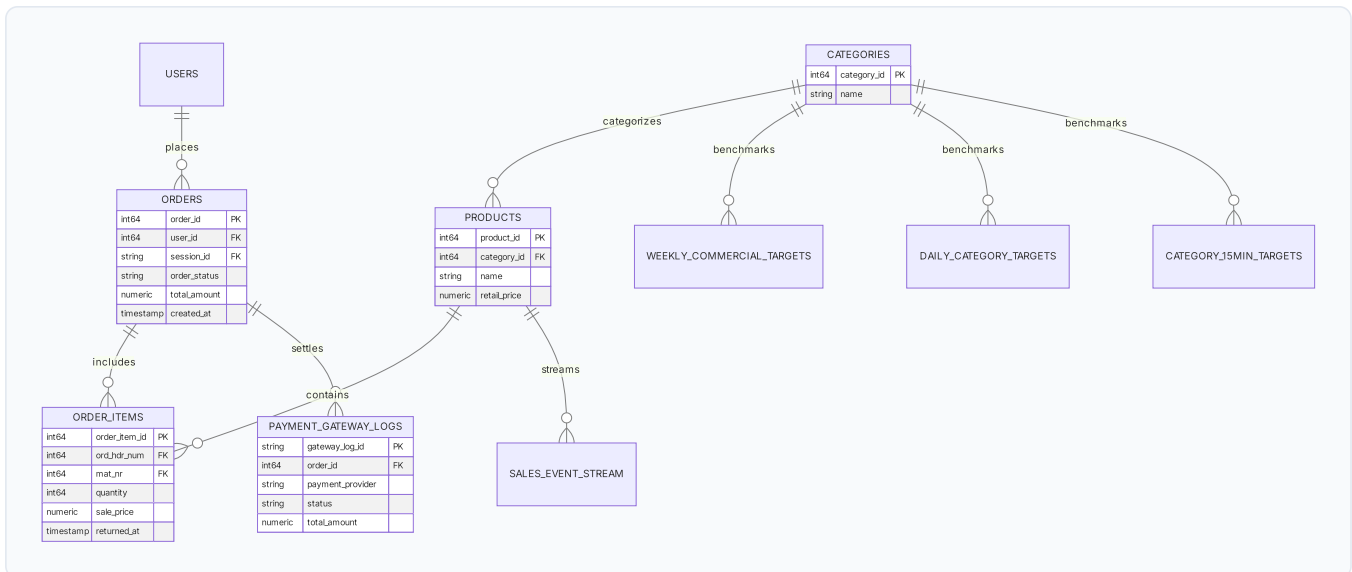
Table Name	Record Count	Data Size	Primary Key	Operational Role & Domain
return_reasons_lookup	5	0.2 KB	*(Surrogate)*	[PURPOSE]: Standardized catalog of customer return reason codes, policy categories, and evidence requirements
stg_adyen_settlements_raw	5	0.3 KB	*(Surrogate)*	[PURPOSE]: Raw daily settlement payout reports from Adyen payment platform
ticket_categories	5	0.2 KB	*(Surrogate)*	[PURPOSE]: Classification taxonomy of customer support issue types and target resolution SLA targets in minutes
categories	4	0.1 KB	*(Surrogate)*	[PURPOSE]: Master product category taxonomy with hierarchical parent references for merchandising and commercial sales reporting
category_hierarchy_paths	4	0.1 KB	*(Surrogate)*	[PURPOSE]: Materialized category hierarchy breadcrumb paths for fast e-commerce storefront navigation and facet filtering
competitor_promotions	4	0.4 KB	*(Surrogate)*	[PURPOSE]: Scraped competitor retail promotional campaigns, discount depths, promotional banners, and relative price parity benchmarking data
discount_coupons_master	4	0.3 KB	*(Surrogate)*	[PURPOSE]: Marketing promotional coupon codes, percentage discount values, minimum order thresholds, and expiration dates
loyalty_tier_definitions	4	0.1 KB	*(Surrogate)*	[PURPOSE]: Loyalty program tier qualification rules, annual spend thresholds in EUR, and promotional point earning multipliers
marketing_campaigns	4	0.3 KB	*(Surrogate)*	[PURPOSE]: Paid digital marketing campaign configurations across Meta Ads and Google Ads with target ROAS, target CPA, and bidding strategy definitions
payment_gateway_fee_schedules	4	0.2 KB	*(Surrogate)*	[PURPOSE]: Contracted merchant interchange fee percentages and fixed per-transaction processing fees across Stripe, PayPal, and Adyen
physical_store_locations	4	0.3 KB	*(Surrogate)*	[PURPOSE]: Master directory of brick-and-mortar retail stores, address geolocations, floor square meters, and opening status
product_attribute_definitions	4	0.2 KB	*(Surrogate)*	[PURPOSE]: Master Product Information Management (PIM) attribute definitions (e
warehouse_zones	4	0.2 KB	*(Surrogate)*	[PURPOSE]: Warehouse physical floor zones (e
email_campaign_templates	3	0.4 KB	*(Surrogate)*	[PURPOSE]: Marketing email templates, subject line A/B test variations, and creative layouts for automated CRM lifecycle journeys
freight_carrier_contracts	3	0.2 KB	*(Surrogate)*	[PURPOSE]: Master freight carrier commercial service contracts, contracted base rates per kilogram, and fuel surcharge formulas

Table Name	Record Count	Data Size	Primary Key	Operational Role & Domain
influencer_campaigns	3	0.3 KB	*(Surrogate)*	[PURPOSE]: Creator marketing performance tracking sponsored content views, promo code redemptions, target vs actual revenue quotas, and creator fees
product_brand_guidelines	3	0.1 KB	*(Surrogate)*	[PURPOSE]: Manufacturer minimum advertised price (MAP) policies, brand authorization rules, and trademark guidelines
product_size_charts	3	0.1 KB	*(Surrogate)*	[PURPOSE]: Category size conversion tables, body measurements in centimeters, and international size mapping
distribution_centers	2	0.1 KB	*(Surrogate)*	[PURPOSE]: Regional fulfillment centers and warehouse logistics hubs managing physical inventory and order dispatch across Europe
intercompany_transfer_pricing	2	0.2 KB	*(Surrogate)*	[PURPOSE]: Cross-border intercompany transfer pricing schedules and cost-plus markup percentages between European corporate subsidiaries
mobile_app_push_campaigns	2	0.2 KB	*(Surrogate)*	[PURPOSE]: Mobile app push notification broadcasts, rich media titles, and deep link targets for iOS/Android apps
seo_meta_tags_registry	2	0.4 KB	*(Surrogate)*	[PURPOSE]: Search engine optimization (SEO) title tags, meta descriptions, and canonical URLs across all web catalog pages
sms_marketing_broadcasts	2	0.3 KB	*(Surrogate)*	[PURPOSE]: SMS text marketing campaign broadcasts, promotional message copy, and targeted customer cohort segments

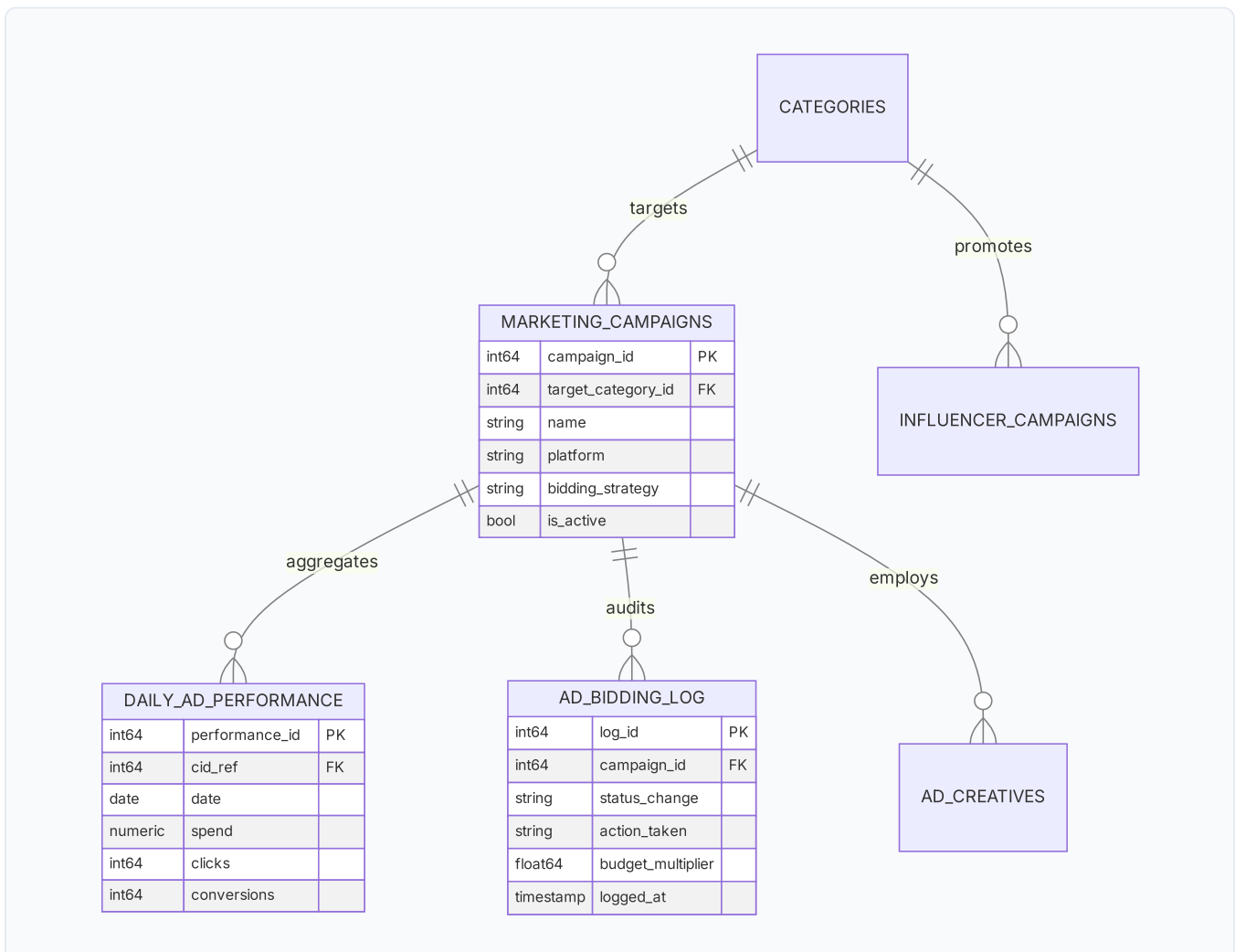
2. Core Relational Architecture Diagrams

The following Mermaid diagrams illustrate the physical foreign key linkages across the core operational clusters:

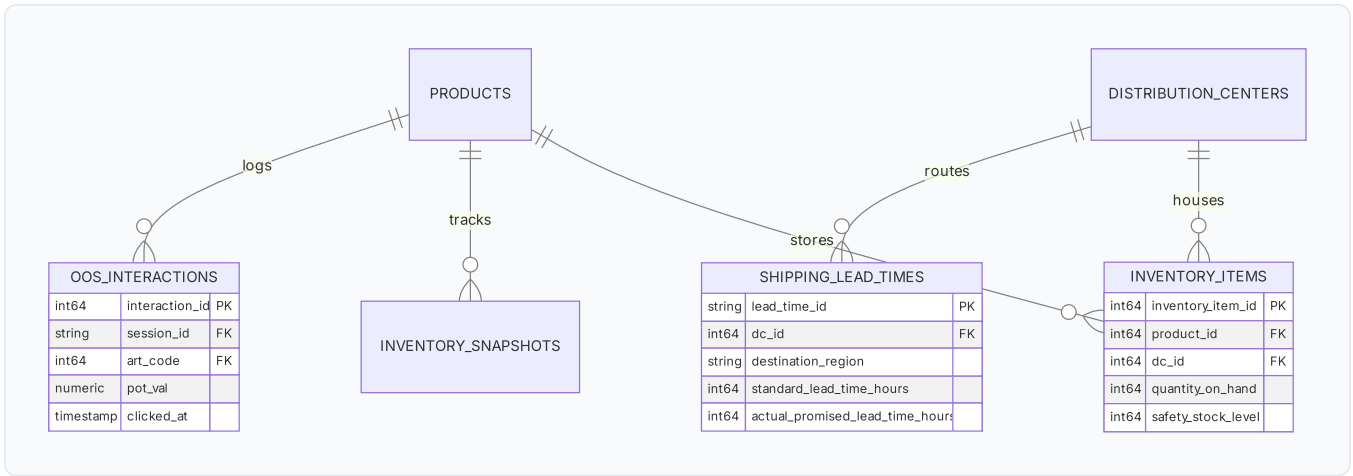
A. Commercial Revenue & Order Transaction Flow



B. Paid Advertising & Automated Bidding Engine



C. Supply Chain Stockouts & Fulfillment



3. Core Investigation Tables Deep-Dive (25 Tables)

Detailed column schemas, primary keys, foreign keys, and exact record counts for all 25 tables in the core diagnostic cluster:

weekly_commercial_targets

- **Business Meaning:** [PURPOSE]: Executive commercial financial planning benchmarks, expected visitor sessions, target revenue in EUR, and conversion rate (CVR) goals by product category and calendar week for sales target tracking and commercial performance analysis. [DOMAIN]: Domain B: Commercial Pacing Targets. [GRAIN]: One row per product category per calendar week (target_id). [TIER & REFRESH]: GOLD_CURATED | Pre-Season Financial Benchmark. [ANALYTICAL ROLE]: Commercial Planning - Weekly Commercial Quotas & Financial Benchmark Targets.
- **Record Count:** 28 rows (1.8 KB)
- **Primary Key (PK):** category_id + week_start_date
- **Foreign Keys (FK):** target_id → targets.target_id

Column Name	Data Type	Field Meaning & Calculation Formula
category_id	INT64	Foreign key to categories table
target_conversion_rate	FLOAT64	Target e-commerce conversion rate (CVR)
target_id	INT64	Commercial target identifier (Primary Key)
target_revenue	FLOAT64	Total planned commercial revenue target in EUR
target_sessions	INT64	Expected web traffic sessions
week_start_date	DATE	Target week start date (2026-11-23)

daily_category_targets

- **Business Meaning:** [PURPOSE]: Intra-week daily pacing targets, budget allocations, expected conversion rates, and commercial revenue quotas per category for daily sales pacing and commercial performance analysis. [DOMAIN]: Domain B: Commercial Pacing Targets. [GRAIN]: One row per category per calendar day (target_id). [TIER & REFRESH]: GOLD_CURATED | Daily Benchmark. [ANALYTICAL ROLE]: Commercial Planning - Daily Revenue Quotas, Planned Conversion Rates & Pacing Baselines.
- **Record Count:** 200 rows (24.4 KB)
- **Primary Key (PK):** category_id + date
- **Foreign Keys (FK):** target_id → targets.target_id

Column Name	Data Type	Field Meaning & Calculation Formula
category_id	INT64	Foreign key to categories table
date	DATE	Target calendar date (2026-11-23 to 2026-11-30)
target_ad_spend	FLOAT64	Allocated paid marketing ad spend budget in EUR
target_aov	FLOAT64	Expected average order value in EUR
target_conversion_rate	FLOAT64	Expected conversion rate benchmark
target_id	STRING	Daily target identifier (Primary Key)
target_revenue	FLOAT64	Daily planned category revenue in EUR
target_roas	FLOAT64	Target return on ad spend multiplier benchmark
target_sessions	INT64	Daily planned web session traffic target

 category_15min_targets

- **Business Meaning:** [PURPOSE]: Intraday 15-minute pacing targets modeling hourly customer traffic waves, intake velocity benchmarks, and intra-day pacing curves. [DOMAIN]: Domain B: Commercial Pacing Targets. [GRAIN]: One row per category per 15-minute time window (target_id). [TIER & REFRESH]: GOLD_CURATED | Intraday Curve Benchmark. [ANALYTICAL ROLE]: Intraday Pacing - 15-Minute Intake Velocity & Demand Curves.
- **Record Count:** 2,688 rows (162.8 KB)
- **Primary Key (PK):** category_id + interval_start_time
- **Foreign Keys (FK):** target_id → targets.target_id

Column Name	Data Type	Field Meaning & Calculation Formula
category_id	INT64	Foreign key to categories table
day_of_week	INT64	Day of week integer (1=Sunday..7=Saturday)
target_id	STRING	15-minute pacing target identifier (Primary Key)
target_revenue	FLOAT64	Planned revenue target for the 15-minute interval in EUR
target_sessions	INT64	Planned web sessions traffic for the interval
time_bucket	TIME	15-minute time bucket

 categories

- **Business Meaning:** [PURPOSE]: Master product category taxonomy with hierarchical parent references for merchandising and commercial sales reporting. [DOMAIN]: Domain A: Core Commerce Catalog. [GRAIN]: One row per product category (category_id). [TIER & REFRESH]: GOLD_CURATED | Batch Daily @ 01:00 UTC. [ANALYTICAL ROLE]: Catalog Dimension - Category Hierarchy & Merchandising Taxonomy.
- **Record Count:** 4 rows (0.1 KB)
- **Primary Key (PK):** None
- **Foreign Keys (FK):** category_id → categories.category_id , parent_category_id → parent_categorys.parent_category_id

Column Name	Data Type	Field Meaning & Calculation Formula
category_id	INT64	Unique identifier for the product category (Primary Key)
name	STRING	Display name of the category (e.g., Beauty, Electronics, Fashion, Home)
parent_category_id	STRING	Self-referencing link to parent category for hierarchy
slug	STRING	URL-safe slug text string

 products

- **Business Meaning:** [PURPOSE]: Master catalog of all retail products, default selling prices, cost of goods, brands, and category mappings for commercial intake tracking. [DOMAIN]: Domain A: Core Commerce Catalog. [GRAIN]: One row per distinct product

SKU (product_id). [TIER & REFRESH]: GOLD_CURATED | Batch Daily @ 01:00 UTC. [ANALYTICAL ROLE]: Product Dimension - SKU Metadata, Pricing & Unit Economics.

- **Record Count:** 600 rows (55.1 KB)
- **Primary Key (PK):** product_id
- **Foreign Keys (FK):** category_id → categories.category_id

Column Name	Data Type	Field Meaning & Calculation Formula
brand	STRING	Brand or manufacturer name
category_id	INT64	Foreign key to categories table
cost	FLOAT64	Acquisition and production cost in EUR
is_active	BOOL	Catalog visibility flag
name	STRING	Full retail product name
product_id	INT64	Unique master identifier for the product (Primary Key)
retail_price	FLOAT64	Default selling price in EUR
sku	STRING	Stock Keeping Unit code

orders

- **Business Meaning:** [PURPOSE]: Core transactional sales order headers capturing customer checkout completion, gross merchandise value (GMV), order status, payment confirmation, and timestamps for commercial revenue analysis. [DOMAIN]: Domain B: Transactions & Target Curves. [GRAIN]: One row per checkout order header (order_id). [TIER & REFRESH]: GOLD_CURATED | Real-Time Streaming. [ANALYTICAL ROLE]: Sales Intake - Gross Merchandise Value (GMV) & Revenue Realization Reporting.
- **Record Count:** 128,997 rows (13.16 MB)
- **Primary Key (PK):** order_id
- **Foreign Keys (FK):** user_id → users.user_id

Column Name	Data Type	Field Meaning & Calculation Formula
created_at	TIMESTAMP	Transaction execution timestamp
num_of_items	INT64	Total physical items count in order
order_id	INT64	Transaction header identifier (Primary Key)
order_status	STRING	Operational status (Completed, Processing, Cancelled)
shipping_fee	FLOAT64	Shipping delivery fee billed in EUR
tax_amount	FLOAT64	VAT tax portion of transaction in EUR
total_amount	FLOAT64	Total gross purchase price in EUR
user_id	INT64	Foreign key to users table

order_items

- **Business Meaning:** [PURPOSE]: Discrete purchase line items recording product SKU references, realized sale prices, promotional discounts, and basket-level revenue realization. [DOMAIN]: Domain B: Transactions & Target Curves. [GRAIN]: One row per purchased item line (order_item_id). [TIER & REFRESH]: GOLD_CURATED | Real-Time Streaming. [ANALYTICAL ROLE]: Line Item Sales - SKU-Level Contribution, Basket Margins & Volume Breakdown.
- **Record Count:** 207,062 rows (19.97 MB)
- **Primary Key (PK):** ord_hdr_num + order_item_id
- **Foreign Keys (FK):** inventory_item_id → inventory_items.inventory_item_id , mat_nr → products.product_id , user_id → users.user_id

Column Name	Data Type	Field Meaning & Calculation Formula
created_at	TIMESTAMP	Line item creation timestamp
delivered_at	TIMESTAMP	Customer delivery confirmation timestamp
discount_amount	FLOAT64	Promotional discount applied in EUR
inventory_item_id	INT64	Foreign key to inventory_items table
mat_nr	INT64	Material article identifier referencing products.product_id
ord_hdr_num	INT64	Foreign key referencing master orders.order_id
order_item_id	INT64	Transaction line item identifier (Primary Key)
quantity	INT64	Quantity of product units purchased
returned_at	TIMESTAMP	Customer return receipt timestamp
sale_price	FLOAT64	Captured unit selling price at checkout in EUR
shipped_at	TIMESTAMP	Logistics carrier dispatch timestamp
user_id	INT64	Foreign key to users table

 sales_event_stream

- **Business Meaning:** [PURPOSE]: Real-time streaming transactional event feed capturing high-frequency sales events and intra-hour intake velocity. [DOMAIN]: Domain B: Transactions & Target Curves. [GRAIN]: One row per real-time purchase event (event_id). [TIER & REFRESH]: GOLD_CURATED | Streaming Event-Driven (<1s). [ANALYTICAL ROLE]: Streaming Velocity - Real-Time Sales Rate & Intra-Hour Order Volume Monitoring.
- **Record Count:** 207,062 rows (21.72 MB)
- **Primary Key (PK):** event_id
- **Foreign Keys (FK):** category_id → categories.category_id , order_id → orders.order_id , product_id → products.product_id

Column Name	Data Type	Field Meaning & Calculation Formula
category_id	INT64	Foreign key to categories table
discount_amount	FLOAT64	Promotional discount deducted in EUR
event_id	STRING	Streaming event UUID identifier (Primary Key)
order_id	INT64	Foreign key to orders table
product_id	INT64	Foreign key to products table
quantity	INT64	Count of units sold
sale_price	FLOAT64	Captured base unit price in EUR
timestamp	TIMESTAMP	Real-time streaming ingestion timestamp

 inventory_items

- **Business Meaning:** [PURPOSE]: Real-time master stock allocations, safety stock warning thresholds, and warehouse batch availability tracking across distribution centers. [DOMAIN]: Domain A: Core Commerce Catalog. [GRAIN]: One row per inventory allocation batch per hub (inventory_item_id). [TIER & REFRESH]: GOLD_CURATED | Real-Time Micro-batch (5 min). [ANALYTICAL ROLE]: Inventory Balance - Real-Time Stock Buffer & Safety Stock Level Tracking.
- **Record Count:** 804 rows (37.7 KB)
- **Primary Key (PK):** inventory_item_id
- **Foreign Keys (FK):** dc_id → dcs.dc_id , product_id → products.product_id

Column Name	Data Type	Field Meaning & Calculation Formula
created_at	TIMESTAMP	Batch intake timestamp
dc_id	INT64	Foreign key to distribution_centers table
inventory_item_id	INT64	Unique identifier for stock batch (Primary Key)
product_id	INT64	Foreign key to products table
quantity_on_hand	INT64	Physical units currently available in stock
safety_stock_level	INT64	Safety stock threshold for reorder alerts

 inventory_snapshots

- **Business Meaning:** [PURPOSE]: Daily and hourly historical inventory snapshots tracking stock depletion, stockouts, and inventory availability across warehouse hubs. [DOMAIN]: Domain A: Core Commerce Catalog. [GRAIN]: One row per SKU per recording timestamp (snapshot_id). [TIER & REFRESH]: GOLD_CURATED | Hourly Snapshot. [ANALYTICAL ROLE]: Inventory Availability - Historical Stock Depletion & Zero-Stock Duration Tracking.
- **Record Count:** **43,800 rows** (1.38 MB)
- **Primary Key (PK):** product_id + distribution_center_id + snapshot_hour
- **Foreign Keys (FK):** snapshot_id → snapshots.snapshot_id

Column Name	Data Type	Field Meaning & Calculation Formula
is_out_of_stock	BOOL	Boolean flag indicating zero stock quantity
product_id	INT64	Foreign key to products table
recorded_at	TIMESTAMP	Snapshot capture timestamp
snapshot_id	INT64	System tracking snapshot identifier (Primary Key)
stock_quantity	INT64	Units remaining at snapshot time

 oos_interactions

- **Business Meaning:** [PURPOSE]: Customer browsing and cart interactions on out-of-stock items, capturing unfulfilled demand and estimated lost revenue in EUR due to inventory stockouts. [DOMAIN]: Domain C: Out-of-Stock Telemetry. [GRAIN]: One row per out-of-stock user click interaction (interaction_id). [TIER & REFRESH]: GOLD_CURATED | Real-Time Event Feed. [ANALYTICAL ROLE]: Lost Demand Telemetry - Stockout Click Tracking & Unfulfilled Demand Estimation.
- **Record Count:** **49,960 rows** (2.51 MB)
- **Primary Key (PK):** interaction_id
- **Foreign Keys (FK):** art_code → products.product_id , session_id → sessions.session_id

Column Name	Data Type	Field Meaning & Calculation Formula
art_code	INT64	Catalog article SKU code referencing products.product_id
clicked_at	TIMESTAMP	Interaction timestamp
interaction_id	INT64	Out of stock interaction identifier (Primary Key)
pot_val	FLOAT64	Estimated unrealized sales value from out-of-stock user friction
session_id	STRING	Foreign key to web_sessions table

 distribution_centers

- **Business Meaning:** [PURPOSE]: Regional fulfillment centers and warehouse logistics hubs managing physical inventory and order dispatch across Europe. [DOMAIN]: Domain A: Core Commerce Catalog. [GRAIN]: One row per logistics hub (dc_id). [TIER & REFRESH]: GOLD_CURATED | Master Static. [ANALYTICAL ROLE]: Fulfillment Dimension - Regional Logistics Hub Partitioning & Dispatch Routing.
- **Record Count:** **5 rows** (0.2 KB)
- **Primary Key (PK):** None

- **Foreign Keys (FK):** dc_id → dcs.dc_id

Column Name	Data Type	Field Meaning & Calculation Formula
dc_id	INT64	Unique identifier for the logistics hub (Primary Key)
latitude	FLOAT64	Hub geolocation latitude
longitude	FLOAT64	Hub geolocation longitude
name	STRING	Logistics hub name (e.g. Paris Nord Fulfilment Centre, Rotterdam Port Hub)

marketing_campaigns

- **Business Meaning:** [PURPOSE]: Paid digital marketing campaign configurations across Meta Ads and Google Ads with target ROAS, target CPA, and bidding strategy definitions. [DOMAIN]: Domain E: Paid Advertising & Attribution. [GRAIN]: One row per advertising campaign (campaign_id). [TIER & REFRESH]: GOLD_CURATED | Batch Hourly Sync. [ANALYTICAL ROLE]: Marketing Directory - Paid Search & Social Campaign Metadata Master.
- **Record Count:** 7 rows (0.5 KB)
- **Primary Key (PK):** None
- **Foreign Keys (FK):** campaign_id → marketing_campaigns.campaign_id, target_category_id → target_categorys.target_category_id

Column Name	Data Type	Field Meaning & Calculation Formula
bidding_strategy	STRING	Automated bidding algorithm (Target ROAS, Maximize Conversions, Manual CPC)
campaign_id	INT64	Marketing campaign identifier (Primary Key)
is_active	BOOL	Boolean flag indicating whether the paid acquisition campaign is actively serving ad impressions.
name	STRING	Campaign display name
platform	STRING	Advertising platform (Meta Ads, Google Ads, TikTok)
target_category_id	INT64	Primary retail category ID targeted by the advertising campaign.

daily_ad_performance

- **Business Meaning:** [PURPOSE]: Daily marketing performance tracking impressions, clicks, advertising spend in EUR, attributed conversions, average CPC, and ROAS efficiency. [DOMAIN]: Domain E: Ad Spend & Paid Traffic. [GRAIN]: One row per campaign per calendar day (perf_id). [TIER & REFRESH]: GOLD_CURATED | Batch Daily @ 03:00 UTC. [ANALYTICAL ROLE]: Paid Acquisition - Daily Spend Velocity, CPC Efficiency & Attributed ROAS Tracking.
- **Record Count:** 188 rows (14.7 KB)
- **Primary Key (PK):** cid_ref + date
- **Foreign Keys (FK):** performance_id → performances.performance_id

Column Name	Data Type	Field Meaning & Calculation Formula
average_cpc	FLOAT64	Average cost per click in EUR
cid_ref	INT64	External marketing campaign identifier referencing marketing_campaigns.campaign_id
clicks	INT64	Total ad clicks generated
conversions	INT64	Total attributed order conversions count
date	DATE	Calendar tracking date (2026-11-23 to 2026-11-27)
impressions	INT64	Total ad impressions served
performance_id	INT64	Primary key surrogate identifier for daily campaign performance record.
spend	FLOAT64	Total advertising expenditure in EUR

ad_bidding_log

- **Business Meaning:** [PURPOSE]: Automated ad platform bidding engine telemetry capturing target ROAS constraints, budget throttling events, and algorithmic learning phase status. [DOMAIN]: Domain E: Bidding Engine Telemetry. [GRAIN]: One row per bidding algorithm throttle/adjustment event (log_id). [TIER & REFRESH]: GOLD_CURATED | Real-Time Platform Webhook. [ANALYTICAL ROLE]: Bidding Telemetry - Automated Target ROAS Constraints & Budget Throttling Telemetry.
- **Record Count:** 46 rows (3.9 KB)
- **Primary Key (PK):** log_id
- **Foreign Keys (FK):** campaign_id → marketing_campaigns.campaign_id

Column Name	Data Type	Field Meaning & Calculation Formula
campaign_id	INT64	Foreign key to marketing_campaigns table
log_id	INT64	Bidding telemetry log identifier (Primary Key)
logged_at	TIMESTAMP	UTC timestamp when the bidding algorithm state change or audit event was logged.
status_change	STRING	Lifecycle status of the automated bidding engine (e.g. LEARNING_COMPLETE, BUDGET_NORMAL, TARGET_ROAS_BREACH_THROTTLED).
action_taken	STRING	Action the bidding engine performed in response to the state change.
observed_cvr_7d	FLOAT64	Trailing seven-day conversion rate the engine observed when it acted.
target_roas_multiplier	FLOAT64	Target return-on-ad-spend constraint in force at the time of the event.
budget_multiplier	FLOAT64	Fraction of nominal daily budget the engine allowed. Below 1.0 means budget was throttled.
bid_adjustment_pct	FLOAT64	Percentage bid adjustment applied alongside the budget change.

ad_creatives

- **Business Meaning:** [PURPOSE]: Creative asset performance tracking creative fatigue, quality scores, click-through rates, and learning-limited states impacting ad delivery. [DOMAIN]: Domain E: Creative Fatigue & Quality Scores. [GRAIN]: One row per creative visual asset (creative_id). [TIER & REFRESH]: GOLD_CURATED | Batch Daily @ 04:00 UTC. [ANALYTICAL ROLE]: Creative Performance - Asset Fatigue, Quality Scores & Algorithmic Delivery Status.
- **Record Count:** 17 rows (1.4 KB)
- **Primary Key (PK):** None
- **Foreign Keys (FK):** parent_adgroup_id → marketing_campaigns.campaign_id , creative_id → creatives.creative_id

Column Name	Data Type	Field Meaning & Calculation Formula
ad_format	STRING	Creative format (Video, Carousel, Static Image)
creative_id	INT64	Creative asset identifier (Primary Key)
dlv_lrn_lmt_flg	BOOL	Boolean flag indicating algorithmic delivery bottleneck (delivery learning-limited)
last_refreshed_at	TIMESTAMP	Timestamp when creative asset was last updated
name	STRING	Creative asset name
parent_adgroup_id	INT64	Parent marketing campaign hierarchy identifier referencing marketing_campaigns.campaign_id
quality_score	INT64	Ad platform quality score (1 to 10 scale)
relevance_status	STRING	Relevance status (ACTIVE, FATIGUED, LOW_QUALITY)

influencer_campaigns

- **Business Meaning:** [PURPOSE]: Creator marketing performance tracking sponsored content views, promo code redemptions, target vs actual revenue quotas, and creator fees. [DOMAIN]: Domain F: Creator & Influencer Marketing. [GRAIN]: One row per creator partnership campaign (campaign_id). [TIER & REFRESH]: GOLD_CURATED | Batch Daily @ 05:00 UTC. [ANALYTICAL ROLE]: Creator Marketing - Influencer Campaign Redemptions & Commission Tracking.
- **Record Count:** 8 rows (1.2 KB)
- **Primary Key (PK):** None
- **Foreign Keys (FK):** influencer_id → influencers.influencer_id

Column Name	Data Type	Field Meaning & Calculation Formula
actual_revenue	FLOAT64	Total gross merchandise value realized directly through the creator unique referral code.
campaign_name	STRING	Marketing campaign title associated with the creator collaboration.
created_at	TIMESTAMP	UTC timestamp when the influencer partnership agreement was activated.
creator_name	STRING	Public social media creator or influencer handle.
fee_amount	FLOAT64	Fixed sponsorship fee in EUR paid to the creator for the promotional campaign.
influencer_id	INT64	Primary key identifier for the creator profile.
is_active	BOOL	Boolean flag indicating whether the creator promotional code is currently valid.
orders_count	INT64	Total number of completed customer checkouts using the creator promotional code.
platform	STRING	Creator platform (Instagram, TikTok, YouTube)
promo_code	STRING	Unique customer discount code assigned to the creator for attribution.
target_revenue	FLOAT64	Contractual commercial target revenue in EUR
views_count	INT64	Total verified social media post/video impressions generated by the creator content.

competitor_price_feed

- **Business Meaning:** [PURPOSE]: Daily competitor retail pricing scrapes benchmarking price parity indices, market elasticity, and competitor price movements. [DOMAIN]: Domain D: Competitor Pricing Intelligence. [GRAIN]: One row per product SKU per competitor scrape (feed_id). [TIER & REFRESH]: GOLD_CURATED | Batch Daily @ 06:00 UTC. [ANALYTICAL ROLE]: Pricing Intelligence - Competitor Price Parity & Elasticity Tracking.
- **Record Count:** 1,200 rows (62.9 KB)
- **Primary Key (PK):** None
- **Foreign Keys (FK):** product_id → products.product_id , scrape_id → scrapes.scrape_id

Column Name	Data Type	Field Meaning & Calculation Formula
competitor_name	STRING	Competitor retail brand (e.g. SephoraEU, DouglasDE, LookFantastic)
competitor_price	FLOAT64	Competitor retail selling price in EUR
is_in_stock	BOOL	Stock availability status of the product on the competitor retail platform.
product_id	INT64	Foreign key to products table
scrape_id	INT64	Unique identifier of the automated market price crawl batch.
scraped_at	TIMESTAMP	Feed scraping timestamp

competitor_promotions

- **Business Meaning:** [PURPOSE]: Scraped competitor retail promotional campaigns, discount depths, promotional banners, and relative price parity benchmarking data. [DOMAIN]: Domain D: Competitor Campaign Intelligence. [GRAIN]: One row per competitor promotion campaign (promo_id). [TIER & REFRESH]: GOLD_CURATED | Batch Daily @ 06:00 UTC. [ANALYTICAL ROLE]: Promotional Intelligence - Competitor Campaign Monitoring & Discount Tracking.
- **Record Count:** 12 rows (1.1 KB)
- **Primary Key (PK):** None
- **Foreign Keys (FK):** category_id → categories.category_id , promo_id → promos.promo_id

Column Name	Data Type	Field Meaning & Calculation Formula
category_id	INT64	Foreign key to categories table
competitor_name	STRING	Competitor brand name
discount_pct	FLOAT64	Promotional discount percentage (e.g. 20.0%)
end_date	DATE	Scheduled expiration date of the competitor promotional discount campaign.
price_index_vs_lumiere	FLOAT64	Price parity ratio comparing competitor retail price against Lumiere standard price (index < 1.0 indicates competitor discounting).
promo_id	INT64	Competitor promotion identifier (Primary Key)
promotion_title	STRING	Public marketing title or discount banner text used in the competitor campaign.
scraped_at	TIMESTAMP	UTC timestamp when the promotional crawl record was captured.
start_date	DATE	Scheduled launch date of the competitor promotional discount campaign.

catalog_recommender_logs

- **Business Meaning:** [PURPOSE]: On-page product recommendation widget impressions capturing algorithm fallback events, category mismatch errors, and lost substitution sales. [DOMAIN]: Domain F: Recommender Engine Telemetry. [GRAIN]: One row per widget recommendation display event (log_id). [TIER & REFRESH]: GOLD_CURATED | Real-Time Clickstream Telemetry. [ANALYTICAL ROLE]: Recommender Telemetry - Algorithmic Fallback Rates & Cross-Sell Engagement.
- **Record Count:** 106,175 rows (9.01 MB)
- **Primary Key (PK):** session_id
- **Foreign Keys (FK):** log_id → logs.log_id , page_category_id → page_categorys.page_category_id , rec_sku → products.product_id , recommended_category_id → recommended_categorys.recommended_category_id , src_sku → products.product_id

Column Name	Data Type	Field Meaning & Calculation Formula
cat_mismatch_flg	INT64	Cross-department mismatch flag: 1 = taxonomy parity error (e.g. Beauty displaying Electronics), 0 = normal
fb_rule_id	INT64	Rule engine identifier: 99 = global category fallback, 0 = standard collaborative filter
log_id	STRING	Recommendation impression log identifier (Primary Key)
opp_cost_eur	NUMERIC	Estimated lost substitution revenue from failed recommendation cross-sell
page_category_id	INT64	Product category ID of the page the customer was viewing when recommendations were served.
rec_sku	INT64	Recommended product candidate suggested by ML engine (FK to products)
recommended_category_id	INT64	Product category ID of the item recommended by the personalization algorithm (used to detect category mismatch fallbacks).
recorded_at	TIMESTAMP	Impression timestamp
session_id	STRING	Foreign key to web_sessions table
src_sku	INT64	Source product item viewed on active page (FK to products)
user_action	STRING	Visitor action (BOUNCED, CLICKED, IGNORED)

shipping_lead_times

- **Business Meaning:** [PURPOSE]: Fulfillment center operational metrics, carrier workload, promised delivery SLAs, and delivery delay impacts on cart abandonment and conversion. [DOMAIN]: Domain F: Fulfillment & Delivery SLAs. [GRAIN]: One row per carrier per destination region per calendar day (lead_time_id). [TIER & REFRESH]: GOLD_CURATED | Batch Daily @ 05:00 UTC. [ANALYTICAL ROLE]: Fulfillment Reliability - Carrier Transit Lead Times & Delivery SLA Tracking.
- **Record Count:** 209 rows (21.3 KB)
- **Primary Key (PK):** None
- **Foreign Keys (FK):** dc_id → dcs.dc_id , lead_time_id → lead_times.lead_time_id

Column Name	Data Type	Field Meaning & Calculation Formula
actual_promised_lead_time_hours	INT64	Real-time checkout promised delivery SLA in hours (e.g. 48h)
capacity_utilization_pct	FLOAT64	Warehouse and carrier capacity utilization percentage
carrier_name	STRING	Logistics carrier (DHL, Chronopost, DPD)
cart_abandonment_impact_pct	FLOAT64	Attributed checkout abandonment percentage from SLA breach
date	DATE	Operational date (2026-11-23 to 2026-11-27)
dc_id	INT64	Foreign key to distribution_centers table
destination_region	STRING	Destination delivery country (France, DACH, Benelux)
estimated_lost_revenue	FLOAT64	Estimated lost checkout revenue in EUR
lead_time_id	STRING	Lead time operational record identifier (Primary Key)
standard_lead_time_hours	INT64	Standard promised delivery SLA in hours (e.g. 24h)

 payment_gateway_logs

- **Business Meaning:** [PURPOSE]: Payment service provider (PSP) authorization logs capturing transaction success rates, latency in milliseconds, error codes, and checkout drop-offs across Stripe, PayPal, and Adyen. [DOMAIN]: Domain F: Payment Gateway Processing. [GRAIN]: One row per PSP gateway transaction attempt (log_id). [TIER & REFRESH]: GOLD_CURATED | Real-Time Streaming Gateway Logs. [ANALYTICAL ROLE]: Checkout Reliability - Payment Gateway Processing Health & Latency Monitoring.
- **Record Count:** 131,283 rows (14.52 MB)
- **Primary Key (PK):** session_id
- **Foreign Keys (FK):** gateway_log_id → gateway_logs.gateway_log_id , order_id → orders.order_id

Column Name	Data Type	Field Meaning & Calculation Formula
country	STRING	Customer billing country localized from IP and checkout address.
created_at	TIMESTAMP	Gateway transaction timestamp
error_code	STRING	Gateway error code (e.g. HTTP_504_GATEWAY_TIMEOUT, CARD_DECLINED)
gateway_log_id	STRING	Unique transaction log identifier issued by the payment gateway.
http_status_code	INT64	HTTP network response code returned by the payment processor API.
latency_ms	INT64	Authorization latency in milliseconds
order_id	INT64	Foreign key to orders table
payment_method	STRING	Customer payment instrument used (e.g. credit_card, paypal_wallet, apple_pay, ideal, klarna).
payment_provider	STRING	Third-party payment service provider (e.g. Stripe, PayPal, Adyen).
session_id	STRING	Web session identifier associated with the checkout transaction attempt.
status	STRING	Transaction status (SUCCESS, FAILED, TIMEOUT)
total_amount	FLOAT64	Gross transaction amount in EUR attempted at payment processing.

 web_sessions

- **Business Meaning:** [PURPOSE]: Web clickstream traffic sessions recording marketing channels, UTM acquisition tags, device operating systems, browsers, and user journey attribution. [DOMAIN]: Domain C: Web Clickstream & Funnel. [GRAIN]: One row per browser session (session_id). [TIER & REFRESH]: GOLD_CURATED | Real-Time Streaming Sessionization. [ANALYTICAL ROLE]: Web Traffic - Acquisition Channel Sessions & Conversion Rate Attribution.
- **Record Count:** 4,270,393 rows (470.06 MB)
- **Primary Key (PK):** session_id
- **Foreign Keys (FK):** user_id → users.user_id

Column Name	Data Type	Field Meaning & Calculation Formula
browser	STRING	Client web browser (Safari, Chrome, Firefox, Edge)
device_os	STRING	Client operating system (iOS, Android, Windows, macOS)
session_id	STRING	Unique browser session UUID (Primary Key)
session_started_at	TIMESTAMP	Session start timestamp
traffic_source	STRING	Origin channel (Paid Search, Paid Social, Direct, Email, Organic)
user_id	INT64	Customer profile identifier (nullable)
utm_campaign	STRING	UTM campaign tag
utm_medium	STRING	UTM medium tag (cpc, social, email, referral)
utm_source	STRING	UTM campaign source tag (google, meta, criteo, newsletter)

 web_events

- **Business Meaning:** [PURPOSE]: User interaction clickstream funnel events capturing product page views, cart additions, checkout initiations, and transaction successes for conversion funnel drop-off analysis. [DOMAIN]: Domain C: Web Clickstream & Funnel. [GRAIN]: One row per clickstream interaction event (event_id). [TIER & REFRESH]: GOLD_CURATED | Real-Time Ingestion Feed. [ANALYTICAL ROLE]: Clickstream Funnel - Micro-Conversions, Page Navigation & Cart Telemetry.
- **Record Count:** 85,605,795 rows (8.05 GB)
- **Primary Key (PK):** session_id
- **Foreign Keys (FK):** event_id → events.event_id , product_id → products.product_id

Column Name	Data Type	Field Meaning & Calculation Formula
created_at	TIMESTAMP	Event interaction timestamp
event_id	INT64	Unique event identifier (Primary Key)
event_type	STRING	Funnel event type (page_view, product_view, cart_add, checkout_start, checkout_success)
metadata	STRUCT<error_message STRING, http_status_code INT64, estimated_lost_revenue NUMERIC>	JSON metadata payload
metadata	STRING	Detailed runtime error string or API gateway failure message.
metadata	NUMERIC	Estimated gross monetary revenue in EUR lost due to the event failure.
metadata	INT64	HTTP network status code recorded at event submission.
page_url	STRING	Page URL path
product_id	INT64	Foreign key to products table (nullable)
session_id	STRING	Foreign key to web_sessions table

 users

- **Business Meaning:** [PURPOSE]: Customer user accounts with demographic localization, country mapping, and lifetime activity profiles. [DOMAIN]: Domain A: Core Commerce Catalog. [GRAIN]: One row per registered customer account (user_id). [TIER & REFRESH]: GOLD_CURATED | Real-Time Ingestion. [ANALYTICAL ROLE]: Customer Dimension - Geographic Cohort Partitioning & Demographic Breakdown.
- **Record Count:** 10,000 rows (1.01 MB)
- **Primary Key (PK):** user_id

Column Name	Data Type	Field Meaning & Calculation Formula
age	INT64	Customer age in years
country	STRING	Customer localization country (e.g. France, Germany)
created_at	TIMESTAMP	Account creation timestamp
email	STRING	Customer contact email address
first_name	STRING	Customer given name
gender	STRING	Self-identified demographic gender
last_name	STRING	Customer surname
latitude	FLOAT64	Customer geolocation coordinate latitude
longitude	FLOAT64	Customer geolocation coordinate longitude
user_id	INT64	Customer account identifier (Primary Key)

4. Extended Enterprise Domains Reference (115 Tables)

The remaining 115 enterprise tables provide deep historical, omnichannel, and operational coverage across 17 functional business domains:

Domain H: Staging & Raw Ingestion (20 Tables)

Table Name	Record Count	Primary Key	Foreign Key Links	Business Description
stg_shopify_orders_raw	2,500	None	payload_id → payloads.payload_id	[PURPOSE]: Raw Shopify webhook JSON payloads capturing e-commerce checkout and order events. [DOMAIN]: Domain H: Staging & Raw Ingestion Tables. [GRAIN]: One row per webhook payload (payload_id). [TIER & REFRESH]: BRONZE_RAW_STAGING
stg_shopify_products_raw	1,200	None	payload_id → payloads.payload_id shopify_product_id → shopify_products.shopify_product_id	[PURPOSE]: Raw Shopify product synchronization JSON payloads from catalog webhooks. [DOMAIN]: Domain H: Staging & Raw Ingestion Tables. [GRAIN]: One row per product payload (payload_id). [TIER & REFRESH]: BRONZE_RAW_STAGING
stg_shopify_customers_raw	1,500	None	payload_id → payloads.payload_id shopify_customer_id → shopify_customers.shopify_customer_id	[PURPOSE]: Raw Shopify customer account update webhook JSON payloads. [DOMAIN]: Domain H: Staging & Raw Ingestion Tables. [GRAIN]: One row per customer payload (payload_id). [TIER & REFRESH]: BRONZE_RAW_STAGING
stg_klaviyo_email_events_raw	5,000	event_id	campaign_id → marketing_campaigns.campaign_id profile_id → profiles.profile_id	[PURPOSE]: Raw event stream from Klaviyo marketing automation webhook (opens, clicks, bounces). [DOMAIN]: Domain H: Staging & Raw Ingestion Tables. [GRAIN]: One row per email event (event_id). [TIER & REFRESH]: BRONZE_RAW_STAGING
stg_klaviyo_campaigns_raw	125	None	campaign_id → marketing_campaigns.campaign_id	[PURPOSE]: Raw email campaign configuration dumps from Klaviyo REST API. [DOMAIN]: Domain H: Staging & Raw Ingestion Tables. [GRAIN]: One row per campaign record (campaign_id). [TIER & REFRESH]: BRONZE_RAW_STAGING
stg_stripe_payment_intents_raw	4,000	None	intent_id → intents.intent_id	[PURPOSE]: Raw Stripe payment intent webhook JSON payloads for credit card

Table Name	Record Count	Primary Key	Foreign Key Links	Business Description
				authorizations. [DOMAIN]: Domain H: Staging & Raw Ingestion Tables. [GRAIN]: One row per payment intent (intent_id). [TIER & REFRESH]: BRONZE_RAW_STAGING
stg_stripe_disputes_raw	300	None	charge_id → charges.charge_id dispute_id → disputes.dispute_id	[PURPOSE]: Raw Stripe credit card chargeback disputes and inquiry logs. [DOMAIN]: Domain H: Staging & Raw Ingestion Tables. [GRAIN]: One row per dispute record (dispute_id). [TIER & REFRESH]: BRONZE_RAW_STAGING
stg_adyen_settlements_raw	5	None	batch_id → batchs.batch_id	[PURPOSE]: Raw daily settlement payout reports from Adyen payment platform. [DOMAIN]: Domain H: Staging & Raw Ingestion Tables. [GRAIN]: One row per settlement batch (batch_id). [TIER & REFRESH]: BRONZE_RAW_STAGING
stg_ga4_clickstream_raw	5,000	None	user_pseudo_id → user_pseudos.user_pseudo_id	[PURPOSE]: Raw Google Analytics 4 export event records from BigQuery streaming export. [DOMAIN]: Domain H: Staging & Raw Ingestion Tables. [GRAIN]: One row per raw GA4 hit event. [TIER & REFRESH]: BRONZE_RAW_STAGING
stg_ga4_traffic_sources_raw	2,500	session_id	(None)	[PURPOSE]: Raw GA4 traffic acquisition attribution source records. [DOMAIN]: Domain H: Staging & Raw Ingestion Tables. [GRAIN]: One row per session traffic acquisition tag (session_id). [TIER & REFRESH]: BRONZE_RAW_STAGING
stg_meta_ad_insights_raw	150	None	adset_id → adsets.adset_id campaign_id → marketing_campaigns.campaign_id	[PURPOSE]: Raw Meta Marketing API adset insights JSON feeds. [DOMAIN]: Domain H: Staging & Raw Ingestion Tables. [GRAIN]: One row per adset reporting interval (adset_id, date_start). [TIER &

Table Name	Record Count	Primary Key	Foreign Key Links	Business Description
				REFRESH]: BRONZE_RAW_STAGING
stg_google_ads_campaigns_raw	125	None	campaign_id → marketing_campaigns.campaign_id	[PURPOSE]: Raw Google Ads campaign performance reporting extracts. [DOMAIN]: Domain H: Staging & Raw Ingestion Tables. [GRAIN]: One row per campaign reporting day (campaign_id, date). [TIER & REFRESH]: BRONZE_RAW_STAGING
stg_google_ads_search_terms_raw	600	None	campaign_id → marketing_campaigns.campaign_id	[PURPOSE]: Raw Google Search Ads query term performance extracts. [DOMAIN]: Domain H: Staging & Raw Ingestion Tables. [GRAIN]: One row per search query per date (search_term, date). [TIER & REFRESH]: BRONZE_RAW_STAGING
stg_criteo_retargeting_raw	100	None	campaign_id → marketing_campaigns.campaign_id	[PURPOSE]: Raw Criteo dynamic product retargeting daily performance reporting. [DOMAIN]: Domain H: Staging & Raw Ingestion Tables. [GRAIN]: One row per retargeting campaign per date (campaign_id, date). [TIER & REFRESH]: BRONZE_RAW_STAGING
stg_wms_shipments_raw	2,000	None	order_id → orders.order_id shipment_id → shipments.shipment_id	[PURPOSE]: Raw warehouse management system (WMS) carrier dispatch manifests. [DOMAIN]: Domain H: Staging & Raw Ingestion Tables. [GRAIN]: One row per warehouse shipment manifest (shipment_id). [TIER & REFRESH]: BRONZE_RAW_STAGING
stg_sap_erp_inventory_feed_raw	1,200	None	batch_id → batches.batch_id plant_id → plants.plant_id	[PURPOSE]: Raw nightly inventory stock feed export from SAP ERP enterprise system. [DOMAIN]: Domain H: Staging & Raw Ingestion Tables. [GRAIN]: One row per SAP material plant balance (batch_id, material_number). [TIER & REFRESH]: BRONZE_RAW_STAGING
stg_sap_erp_purchase_orders_raw	300	None	(None)	[PURPOSE]: Raw SAP ERP procurement

Table Name	Record Count	Primary Key	Foreign Key Links	Business Description
				purchase order dumps. [DOMAIN]: Domain H: Staging & Raw Ingestion Tables. [GRAIN]: One row per SAP purchase order header (po_number). [TIER & REFRESH]: BRONZE_RAW_STAGING
stg_zendesk_tickets_raw	2,500	None	ticket_id → tickets.ticket_id	[PURPOSE]: Raw customer support ticket JSON records extracted from Zendesk Support API. [DOMAIN]: Domain H: Staging & Raw Ingestion Tables. [GRAIN]: One row per ticket record (ticket_id). [TIER & REFRESH]: BRONZE_RAW_STAGING
stg_zendesk_satisfaction_raw	1,500	None	rating_id → ratings.rating_id ticket_id → tickets.ticket_id	[PURPOSE]: Raw customer satisfaction rating survey responses from Zendesk. [DOMAIN]: Domain H: Staging & Raw Ingestion Tables. [GRAIN]: One row per rating response (rating_id). [TIER & REFRESH]: BRONZE_RAW_STAGING
stg_trustpilot_reviews_raw	80	None	review_id → reviews.review_id	[PURPOSE]: Raw Trustpilot public customer review feed webhooks. [DOMAIN]: Domain H: Staging & Raw Ingestion Tables. [GRAIN]: One row per customer review (review_id). [TIER & REFRESH]: BRONZE_RAW_STAGING

Domain I: Returns, Refunds & Reverse Logistics (10 Tables)

Table Name	Record Count	Primary Key	Foreign Key Links	Business Description
product_returns	1,500	None	order_id → orders.order_id order_item_id → order_items.order_item_id	[PURPOSE]: Customer post-delivery return authorizations (RMA), return reasons, processing statuses, and item condition triage for reverse logistics and refund operations. [DOMAIN]: Domain I: Returns, Refunds & RMA Management. [GRAIN]: One row per returned item authorization (return_id). [TIER & REFRESH]: GOLD_CURATED
return_inspections	1,500	None	inspection_id → inspections.inspection_id inspector_id → inspectors.inspector_id	[PURPOSE]: Physical warehouse inspection triage logs evaluating item condition and restocking eligibility for returned goods. [DOMAIN]: Domain I: Returns, Refunds & RMA Management. [GRAIN]: One row per return triage inspection (inspection_id). [TIER & REFRESH]: GOLD_CURATED
return_shipping_labels	1,500	None	label_id → labels.label_id return_id → returns.return_id	[PURPOSE]: Logistics return shipping waybills and tracking records generated for customer merchandise returns. [DOMAIN]: Domain I: Returns, Refunds & RMA Management. [GRAIN]: One row per return parcel label (label_id). [TIER & REFRESH]: SILVER_CONSOLIDATED
return_reasons_lookup	5	None	(None)	[PURPOSE]: Standardized catalog of customer return reason codes, policy categories, and evidence requirements. [DOMAIN]: Domain I: Returns, Refunds & RMA Management. [GRAIN]: One row per return reason code (reason_code). [TIER & REFRESH]: GOLD_CURATED
customer_refunds	1,500	None	gateway_refund_id → gateway_refunds.gateway_refund_id order_id → orders.order_id	[PURPOSE]: Monetary refund disbursements issued to customers following verified returns, cancellations, or damaged item claims. [DOMAIN]: Domain I: Returns, Refunds & RMA Management. [GRAIN]: One row per refund transaction (refund_id). [TIER & REFRESH]: GOLD_CURATED
replacement_orders	300	None	new_order_id → new_orders.new_order_id original_order_id → original_orders.original_order_id	[PURPOSE]: Zero-cost replacement orders dispatched to customers for lost, defective, or incorrect shipments. [DOMAIN]: Domain I: Returns, Refunds & RMA Management. [GRAIN]: One row per replacement order link (replacement_id). [TIER & REFRESH]: GOLD_CURATED
restocking_fee_logs	150	None	fee_id → fees.fee_id return_id → returns.return_id	[PURPOSE]: Restocking fee deductions applied to customer return refunds for out-of-policy returns or bulky freight items. [DOMAIN]: Domain I: Returns, Refunds & RMA Management. [GRAIN]: One row per fee deduction

Table Name	Record Count	Primary Key	Foreign Key Links	Business Description
				(fee_id). [TIER & REFRESH]: SILVER_CONSOLIDATED
warranty_claims	200	None	claim_id → claims.claim_id product_id → products.product_id	[PURPOSE]: Customer warranty claims, defect reports, manufacturer claim filings, and resolution triage. [DOMAIN]: Domain I: Returns, Refunds & RMA Management. [GRAIN]: One row per warranty claim filing (claim_id). [TIER & REFRESH]: GOLD_CURATED
store_credit_issuances	80	None	credit_id → credits.credit_id user_id → users.user_id	[PURPOSE]: Store credit ledger records, customer credit balances, and credit expiration tracking. [DOMAIN]: Domain I: Returns, Refunds & RMA Management. [GRAIN]: One row per store credit grant (credit_id). [TIER & REFRESH]: SILVER_CONSOLIDATED
accounts_payable_disbursements	150	None	disbursement_id → disbursements.disbursement_id invoice_id → invoices.invoice_id	[PURPOSE]: Outbound cash disbursement payments executed to settle vendor accounts payable invoices. [DOMAIN]: Domain L: Finance, General Ledger, Tax & Accounting. [GRAIN]: One row per payment disbursement (disbursement_id). [TIER & REFRESH]: SILVER_CONSOLIDATED

📁 Domain J: Customer Support, CRM & Voice of Customer (12 Tables)

Table Name	Record Count	Primary Key	Foreign Key Links	Business Description
support_tickets	2,000	None	order_id → orders.order_id ticket_id → tickets.ticket_id	[PURPOSE]: Customer service support cases, ticket priority, omnichannel resolution duration, and first-response metrics. [DOMAIN]: Domain J: Customer Support, CRM & CSAT. [GRAIN]: One row per customer service ticket (ticket_id). [TIER & REFRESH]: GOLD_CURATED
ticket_messages	5,000	None	message_id → messages.message_id sender_id → senders.sender_id	[PURPOSE]: Detailed communication thread messages exchanged between support agents and customers. [DOMAIN]: Domain J: Customer Support, CRM & CSAT. [GRAIN]: One row per message in ticket thread (message_id). [TIER & REFRESH]: SILVER_CONSOLIDATED
support_agents	125	None	agent_id → agents.agent_id	[PURPOSE]: Customer support representative profiles, tier rankings, language capabilities, and assigned shift schedules. [DOMAIN]: Domain J: Customer Support, CRM & CSAT. [GRAIN]: One row per support agent (agent_id). [TIER & REFRESH]: GOLD_CURATED
ticket_categories	5	None	category_id → categories.category_id	[PURPOSE]: Classification taxonomy of customer support issue types and target resolution SLA targets in minutes. [DOMAIN]: Domain J: Customer Support, CRM & CSAT. [GRAIN]: One row per ticket category (category_id). [TIER & REFRESH]: GOLD_CURATED
csat_surveys	1,500	None	survey_id → surveys.survey_id ticket_id → tickets.ticket_id	[PURPOSE]: Post-service customer satisfaction survey scores (1-5 scale) and verbatim feedback submitted following ticket resolution. [DOMAIN]: Domain J: Customer Support, CRM & CSAT. [GRAIN]: One row per survey response (survey_id). [TIER & REFRESH]: GOLD_CURATED
nps_feedback_responses	150	None	nps_id → npss.nps_id user_id → users.user_id	[PURPOSE]: Relationship Net Promoter Score (NPS) surveys measuring overall customer brand loyalty and willingness to recommend.

Table Name	Record Count	Primary Key	Foreign Key Links	Business Description
				[DOMAIN]: Domain J: Customer Support, CRM & CSAT. [GRAIN]: One row per NPS survey submission (nps_id). [TIER & REFRESH]: GOLD_CURATED
live_chat_sessions	1,500	None	agent_id → agents.agent_id chat_session_id → chat_sessions.chat_session_id	[PURPOSE]: Real-time live chat sessions between web visitors and support agents, recording queue wait times and chat duration. [DOMAIN]: Domain J: Customer Support, CRM & CSAT. [GRAIN]: One row per live chat session (chat_session_id). [TIER & REFRESH]: SILVER_CONSOLIDATED
live_chat_messages	4,000	None	chat_session_id → chat_sessions.chat_session_id message_id → messages.message_id	[PURPOSE]: Individual real-time chat transcript messages exchanged during live customer chat sessions. [DOMAIN]: Domain J: Customer Support, CRM & CSAT. [GRAIN]: One row per chat transcript message (message_id). [TIER & REFRESH]: SILVER_CONSOLIDATED
customer_escalations	125	None	escalation_id → escalations.escalation_id manager_id → managers.manager_id	[PURPOSE]: High-priority customer escalations handled by management, documenting escalation reason, resolution category, and financial goodwill concessions in EUR. [DOMAIN]: Domain J: Customer Support, CRM & CSAT. [GRAIN]: One row per escalation case (escalation_id). [TIER & REFRESH]: GOLD_CURATED
call_center_recordings_metadata	150	None	call_id → calls.call_id	[PURPOSE]: Inbound telephony call records, interactive voice response (IVR) menu traversal, call duration, and customer phone hashes. [DOMAIN]: Domain J: Customer Support, CRM & CSAT. [GRAIN]: One row per voice telephony call (call_id). [TIER & REFRESH]: SILVER_CONSOLIDATED
knowledge_base_articles	300	None	article_id → articles.article_id	[PURPOSE]: Help center self-service FAQ articles, view counts, and customer helpfulness voting metrics. [DOMAIN]: Domain J: Customer Support, CRM & CSAT. [GRAIN]: One row per

Table Name	Record Count	Primary Key	Foreign Key Links	Business Description
				knowledge base article (article_id). [TIER & REFRESH]: GOLD_CURATED
agent_interaction_logs	852	interaction_id	data_agent_id → data_agents.data_agent_id job_id → jobs.job_id	[PURPOSE]: Conversational Analytics session audit logs recording natural language prompts, generated SQL queries, latency, and token execution costs. [DOMAIN]: Domain G: Agent Governance & Audit. [GRAIN]: One row per conversational user interaction turn (interaction_id). [TIER & REFRESH]: SILVER_CONSOLIDATED

 Domain K: Supply Chain, Procurement & Multi-DC Warehousing (14 Tables)

Table Name	Record Count	Primary Key	Foreign Key Links	Business Description
<code>purchase_orders</code>	60	None	destination_dc_id → <code>destination_dcs.destination_dc_id</code> po_id → <code>pos.po_id</code>	[PURPOSE]: Commercial procurement purchase order headers issued to product suppliers with contractual delivery deadlines and values in EUR. [DOMAIN]: Domain K: Supply Chain, Procurement & Warehousing. [GRAIN]: One row per purchase order header (po_id). [TIER & REFRESH]: GOLD_CURATED
<code>purchase_order_line_items</code>	1,200	None	po_id → <code>pos.po_id</code> po_line_id → <code>po_lines.po_line_id</code>	[PURPOSE]: Itemized procurement line items specifying ordered quantities, received quantities, and unit wholesale costs in EUR. [DOMAIN]: Domain K: Supply Chain, Procurement & Warehousing. [GRAIN]: One row per purchase order line (po_line_id). [TIER & REFRESH]: GOLD_CURATED
<code>suppliers_master</code>	150	None	supplier_id → <code>suppliers.supplier_id</code>	[PURPOSE]: Global product manufacturer and wholesale supplier master profiles, country codes, payment terms, and active contract status. [DOMAIN]: Domain K: Supply Chain, Procurement & Warehousing. [GRAIN]: One row per vendor supplier (supplier_id). [TIER & REFRESH]: GOLD_CURATED
<code>supplier_lead_time_history</code>	600	None	history_id → <code>historys.history_id</code> product_id → <code>products.product_id</code>	[PURPOSE]: Historical supplier manufacturing and transit lead times comparing promised days vs actual delivery dates. [DOMAIN]: Domain K: Supply Chain, Procurement & Warehousing. [GRAIN]: One row per supplier shipment delivery (history_id). [TIER & REFRESH]: SILVER_CONSOLIDATED
<code>supplier_quality_scorecards</code>	150	None	scorecard_id → <code>scorecards.scorecard_id</code> supplier_id → <code>suppliers.supplier_id</code>	[PURPOSE]: Monthly supplier performance evaluation scorecards tracking On-Time In-Full (OTIF %) delivery and defect rates. [DOMAIN]: Domain K: Supply Chain, Procurement & Warehousing. [GRAIN]: One row per supplier per month (scorecard_id). [TIER & REFRESH]: GOLD_CURATED
<code>inbound_dock_appointments</code>	80	None	appointment_id → <code>appointments.appointment_id</code> dc_id → <code>dcs.dc_id</code>	[PURPOSE]: Warehouse inbound freight dock door appointment schedules, carrier arrival timestamps, and trailer unloading durations. [DOMAIN]: Domain K: Supply Chain, Procurement & Warehousing. [GRAIN]: One row per freight dock appointment (appointment_id). [TIER & REFRESH]: SILVER_CONSOLIDATED

Table Name	Record Count	Primary Key	Foreign Key Links	Business Description
cross_dock_transfer_orders	150	None	destination_dc_id → destination_dcs.destination_dc_id product_id → products.product_id	[PURPOSE]: Inter-hub inventory transfer shipments moving stock between regional distribution centers across Europe. [DOMAIN]: Domain K: Supply Chain, Procurement & Warehousing. [GRAIN]: One row per inter-hub transfer shipment (transfer_id). [TIER & REFRESH]: SILVER_CONSOLIDATED
warehouse_zones	4	None	dc_id → dcs.dc_id zone_id → zones.zone_id	[PURPOSE]: Warehouse physical floor zones (e.g. Mezzanine, Cold Storage, High-Bay Racking) and climate control parameters. [DOMAIN]: Domain K: Supply Chain, Procurement & Warehousing. [GRAIN]: One row per warehouse physical zone (zone_id). [TIER & REFRESH]: GOLD_CURATED
warehouse_aisles_and_racks	160	None	bin_id → bins.bin_id zone_id → zones.zone_id	[PURPOSE]: Storage bin locations, aisle coordinates, vertical rack levels, and maximum weight capacity limits. [DOMAIN]: Domain K: Supply Chain, Procurement & Warehousing. [GRAIN]: One row per physical storage bin (bin_id). [TIER & REFRESH]: SILVER_CONSOLIDATED
pallet_inventory_locations	1,500	None	bin_id → bins.bin_id product_id → products.product_id	[PURPOSE]: Real-time pallet License Plate Number (LPN) tracking linking physical pallets to warehouse bin coordinates. [DOMAIN]: Domain K: Supply Chain, Procurement & Warehousing. [GRAIN]: One row per pallet LPN (pallet_lpn). [TIER & REFRESH]: SILVER_CONSOLIDATED
forklift_telemetry_logs	600	None	equipment_id → equipments.equipment_id telemetry_id → telemetries.telemetry_id	[PURPOSE]: Material handling equipment (forklift) IoT telemetry, battery state of charge, odometer readings, and maintenance alerts. [DOMAIN]: Domain K: Supply Chain, Procurement & Warehousing. [GRAIN]: One row per equipment telemetry ping (telemetry_id). [TIER & REFRESH]: SILVER_CONSOLIDATED
warehouse_labor_shifts	60	None	dc_id → dcs.dc_id employee_id → employees.employee_id	[PURPOSE]: Warehouse worker labor shift schedules, picking tasks completed, and units picked per hour for logistics fulfillment. [DOMAIN]: Domain K: Supply Chain, Procurement & Warehousing. [GRAIN]: One row per employee shift (shift_id). [TIER & REFRESH]: SILVER_CONSOLIDATED
freight_carrier_contracts	3	None	contract_id → contracts.contract_id	[PURPOSE]: Master freight carrier commercial service contracts,

Table Name	Record Count	Primary Key	Foreign Key Links	Business Description
				contracted base rates per kilogram, and fuel surcharge formulas. [DOMAIN]: Domain K: Supply Chain, Procurement & Warehousing. [GRAIN]: One row per carrier contract (contract_id). [TIER & REFRESH]: SILVER_CONSOLIDATED
customs_and_duties_declarations	200	None	declaration_id → declarations.declaration_id po_id → pos.po_id	[PURPOSE]: International import customs declarations, Harmonized System (HS) tariff codes, and paid import duty amounts in EUR. [DOMAIN]: Domain K: Supply Chain, Procurement & Warehousing. [GRAIN]: One row per customs import entry (declaration_id). [TIER & REFRESH]: SILVER_CONSOLIDATED

 Domain L: Finance, General Ledger, Tax & Accounting (12 Tables)

Table Name	Record Count	Primary Key	Foreign Key Links	Business Description
chart_of_accounts	10	None	(None)	[PURPOSE]: Master general ledger chart of accounts defining corporate asset, liability, equity, revenue, and expense account codes. [DOMAIN]: Domain L: Finance, General Ledger, Tax & Accounting. [GRAIN]: One row per financial account (account_number). [TIER & REFRESH]: GOLD_CURATED
general_ledger_journal_entries	125	None	journal_id → journals.journal_id	[PURPOSE]: Official enterprise general ledger accounting journal header entries recording double-entry bookkeeping debits and credits for monthly corporate financial closes. [DOMAIN]: Domain L: Finance, General Ledger, Tax & Accounting. [GRAIN]: One row per journal posting header (journal_id). [TIER & REFRESH]: GOLD_CURATED
gl_journal_lines	250	None	journal_id → journals.journal_id line_id → lines.line_id	[PURPOSE]: Granular debit and credit line item postings mapped to chart of accounts codes for general ledger reconciliation. [DOMAIN]: Domain L: Finance, General Ledger, Tax & Accounting. [GRAIN]: One row per accounting journal line (line_id). [TIER & REFRESH]: SILVER_CONSOLIDATED
accounts_payable_invoices	200	None	invoice_id → invoices.invoice_id	[PURPOSE]: Vendor accounts payable (AP) commercial invoices, invoice amounts in EUR, payment due dates, and approval statuses. [DOMAIN]: Domain L: Finance, General Ledger, Tax & Accounting. [GRAIN]: One row per vendor invoice (invoice_id). [TIER & REFRESH]: GOLD_CURATED
accounts_receivable_invoices	100	None	ar_invoice_id → ar_invoices.ar_invoice_id	[PURPOSE]: B2B corporate client billing invoices, accounts receivable balances, credit terms, and payment settlement tracking. [DOMAIN]: Domain L: Finance, General Ledger, Tax & Accounting. [GRAIN]: One row per corporate billing invoice (ar_invoice_id). [TIER & REFRESH]: GOLD_CURATED
bank_account_reconciliation	5	None	reconciliation_id → reconciliations.reconciliation_id	[PURPOSE]: Monthly corporate bank account balance reconciliations comparing statement ending balances against general ledger cash. [DOMAIN]: Domain L: Finance, General Ledger, Tax & Accounting. [GRAIN]: One row per bank account reconciliation period (reconciliation_id). [TIER & REFRESH]: SILVER_CONSOLIDATED
currency_exchange_rates_daily	20	None	(None)	[PURPOSE]: Daily foreign currency exchange rates benchmarking EUR against USD, GBP, CHF, and other major trading currencies. [DOMAIN]: Domain L: Finance, General Ledger,

Table Name	Record Count	Primary Key	Foreign Key Links	Business Description
				Tax & Accounting. [GRAIN]: One row per currency per calendar day (currency_code, date). [TIER & REFRESH]: GOLD_CURATED
vat_tax_jurisdictions	7	None	(None)	[PURPOSE]: Master European Value Added Tax (VAT) rate tables across European destination countries and standard/reduced rate categories. [DOMAIN]: Domain L: Finance, General Ledger, Tax & Accounting. [GRAIN]: One row per country tax jurisdiction (country_code). [TIER & REFRESH]: GOLD_CURATED
vat_period_filing_reports	7	None	filing_id → filings.filing_id	[PURPOSE]: Periodic statutory VAT tax return filings summarizing taxable sales revenue and collected output VAT across European jurisdictions. [DOMAIN]: Domain L: Finance, General Ledger, Tax & Accounting. [GRAIN]: One row per tax return filing period (filing_id). [TIER & REFRESH]: SILVER_CONSOLIDATED
payment_gateway_fee_schedules	4	None	(None)	[PURPOSE]: Contracted merchant interchange fee percentages and fixed per-transaction processing fees across Stripe, PayPal, and Adyen. [DOMAIN]: Domain L: Finance, General Ledger, Tax & Accounting. [GRAIN]: One row per payment gateway schedule (gateway_name, effective_date). [TIER & REFRESH]: SILVER_CONSOLIDATED
intercompany_transfer_pricing	2	None	schedule_id → schedules.schedule_id	[PURPOSE]: Cross-border intercompany transfer pricing schedules and cost-plus markup percentages between European corporate subsidiaries. [DOMAIN]: Domain L: Finance, General Ledger, Tax & Accounting. [GRAIN]: One row per transfer pricing relationship (schedule_id). [TIER & REFRESH]: SILVER_CONSOLIDATED
warehouse_refurbishments	60	None	product_id → products.product_id refurb_id → refurbishments.refurb_id	[PURPOSE]: Refurbishment work orders, technician labor hours, and replacement parts costs for returned open-box electronics. [DOMAIN]: Domain I: Returns, Refunds & RMA Management. [GRAIN]: One row per refurbishing work order (refurb_id). [TIER & REFRESH]: SILVER_CONSOLIDATED

Domain M: Loyalty, Customer Retention & Rewards (10 Tables)

Table Name	Record Count	Primary Key	Foreign Key Links	Business Description
loyalty_members	2,500	None	membership_id → memberships.membership_id user_id → users.user_id	[PURPOSE]: Customer loyalty program membership profiles, reward tier standings (Silver, Gold, Platinum), and active point balances. [DOMAIN]: Domain M: Loyalty, Customer Retention & Rewards. [GRAIN]: One row per loyalty member account (membership_id). [TIER & REFRESH]: GOLD_CURATED
loyalty_points_ledger	4,000	None	ledger_id → ledgers.ledger_id order_id → orders.order_id	[PURPOSE]: Detailed transactional audit ledger recording loyalty points earned from purchases, bonus campaigns, and point expirations. [DOMAIN]: Domain M: Loyalty, Customer Retention & Rewards. [GRAIN]: One row per loyalty point transaction event (ledger_id). [TIER & REFRESH]: SILVER_CONSOLIDATED
loyalty_tier_definitions	4	None	(None)	[PURPOSE]: Loyalty program tier qualification rules, annual spend thresholds in EUR, and promotional point earning multipliers. [DOMAIN]: Domain M: Loyalty, Customer Retention & Rewards. [GRAIN]: One row per loyalty tier level (tier_name). [TIER & REFRESH]: GOLD_CURATED
loyalty_reward_redemptions	600	None	redemption_id → redemptions.redemption_id user_id → users.user_id	[PURPOSE]: Member reward redemption events converting accumulated loyalty points into discount vouchers or merchandise gifts. [DOMAIN]: Domain M: Loyalty, Customer Retention & Rewards. [GRAIN]: One row per reward redemption (redemption_id). [TIER & REFRESH]: SILVER_CONSOLIDATED
gift_card_master	80	None	card_id → cards.card_id purchaser_user_id → purchaser_users.purchaser_user_id	[PURPOSE]: Electronic and physical gift card registry, initial balances in EUR, current remaining balances, and activation states. [DOMAIN]: Domain M: Loyalty, Customer Retention & Rewards. [GRAIN]: One row per issued gift card (card_id). [TIER & REFRESH]: GOLD_CURATED
gift_card_transactions	120	None	card_id → cards.card_id order_id → orders.order_id	[PURPOSE]: Individual debit and top-up transactions charged against customer gift card balances during checkout. [DOMAIN]: Domain M: Loyalty, Customer Retention & Rewards. [GRAIN]: One row per gift card redemption transaction (transaction_id). [TIER & REFRESH]: SILVER_CONSOLIDATED
referral_program_invites	600	None	invite_id → invites.invite_id referrer_user_id → referrer_users.referrer_user_id	[PURPOSE]: Customer referral invitation tracking, unique referral links, and recipient invitation delivery logs. [DOMAIN]: Domain M: Loyalty, Customer Retention & Rewards. [GRAIN]: One row per referral invitation dispatched (invite_id). [TIER & REFRESH]: SILVER_CONSOLIDATED

Table Name	Record Count	Primary Key	Foreign Key Links	Business Description
referral_reward_claims	200	None	claim_id → claims.claim_id invite_id → invites.invite_id	[PURPOSE]: Referral rewards credited to advocating customers upon successful order completion by referred friends. [DOMAIN]: Domain M: Loyalty, Customer Retention & Rewards. [GRAIN]: One row per referral bonus claim (claim_id). [TIER & REFRESH]: SILVER_CONSOLIDATED
user_subscription_preferences	2,500	None	preference_id → preferences.preference_id user_id → users.user_id	[PURPOSE]: Customer communication consent settings, marketing opt-ins (email, SMS, push), and GDPR compliance timestamps. [DOMAIN]: Domain N: Lifecycle Marketing, Email & Push. [GRAIN]: One row per customer communication preference (preference_id). [TIER & REFRESH]: GOLD_CURATED
coupon_redemption_audit	1,500	None	audit_id → audits.audit_id coupon_id → coupons.coupon_id	[PURPOSE]: Audit logs recording every checkout coupon code application, user ID, and monetary discount amount realized in EUR. [DOMAIN]: Domain M: Loyalty, Customer Retention & Rewards. [GRAIN]: One row per checkout coupon application (audit_id). [TIER & REFRESH]: SILVER_CONSOLIDATED

 Domain N: Lifecycle Marketing, Email & Push Telemetry (10 Tables)

Table Name	Record Count	Primary Key	Foreign Key Links	Business Description
email_send_queue_logs	5,000	None	send_id → sends.send_id template_id → templates.template_id	[PURPOSE]: Outbound email dispatch queue execution logs, deliverability statuses, and send timestamps. [DOMAIN]: Domain N: Lifecycle Marketing, Email & Push. [GRAIN]: One row per dispatched email transmission (send_id). [TIER & REFRESH]: SILVER_CONSOLIDATED
email_bounces_and_complaints	150	None	bounce_id → bounces.bounce_id user_id → users.user_id	[PURPOSE]: Hard/soft email bounce records, spam complaints, and invalid email addresses for sender reputation management. [DOMAIN]: Domain N: Lifecycle Marketing, Email & Push. [GRAIN]: One row per bounce or complaint event (bounce_id). [TIER & REFRESH]: SILVER_CONSOLIDATED
email_campaign_templates	3	None	template_id → templates.template_id	[PURPOSE]: Marketing email templates, subject line A/B test variations, and creative layouts for automated CRM lifecycle journeys. [DOMAIN]: Domain N: Lifecycle Marketing, Email & Push. [GRAIN]: One row per email template (template_id). [TIER & REFRESH]: GOLD_CURATED
sms_marketing_broadcasts	2	None	sms_campaign_id → sms_campaigns.sms_campaign_id	[PURPOSE]: SMS text marketing campaign broadcasts, promotional message copy, and targeted customer cohort segments. [DOMAIN]: Domain N: Lifecycle Marketing, Email & Push. [GRAIN]: One row per SMS marketing campaign (sms_campaign_id). [TIER & REFRESH]: SILVER_CONSOLIDATED
sms_delivery_receipts	2,000	None	receipt_id → receipts.receipt_id sms_campaign_id → sms_campaigns.sms_campaign_id	[PURPOSE]: Telecommunications carrier SMS delivery receipts and handset delivery confirmations. [DOMAIN]: Domain N: Lifecycle Marketing, Email & Push. [GRAIN]: One row per SMS message delivery receipt (receipt_id). [TIER & REFRESH]: SILVER_CONSOLIDATED
mobile_app_push_campaigns	2	None	push_campaign_id → push_campaigns.push_campaign_id	[PURPOSE]: Mobile app push notification broadcasts, rich media titles, and deep link targets for iOS/Android apps. [DOMAIN]: Domain N: Lifecycle Marketing, Email & Push. [GRAIN]: One row per push campaign broadcast (push_campaign_id). [TIER & REFRESH]: SILVER_CONSOLIDATED
push_notification_receipts	2,500	None	push_campaign_id → push_campaigns.push_campaign_id push_id → pushes.push_id	[PURPOSE]: Individual mobile app push notification delivery receipts and user tap/click interactions. [DOMAIN]: Domain N: Lifecycle Marketing, Email & Push. [GRAIN]: One row per push receipt event (push_id). [TIER & REFRESH]: SILVER_CONSOLIDATED
affiliate_publishers_directory	100	None	affiliate_id → affiliates.affiliate_id	[PURPOSE]: Third-party affiliate publisher networks, promotional partners, and contracted commission rates. [DOMAIN]: Domain N: Lifecycle Marketing, Email & Push. [GRAIN]: One

Table Name	Record Count	Primary Key	Foreign Key Links	Business Description
				row per affiliate partner (affiliate_id). [TIER & REFRESH]: GOLD_CURATED
affiliate_commission_payouts	100	None	affiliate_id → affiliates.affiliate_id payout_id → payouts.payout_id	[PURPOSE]: Monthly affiliate publisher commission calculations, attributed sales volume in EUR, and payout approval records. [DOMAIN]: Domain N: Lifecycle Marketing, Email & Push. [GRAIN]: One row per affiliate monthly billing statement (payout_id). [TIER & REFRESH]: SILVER_CONSOLIDATED
discount_coupons_master	4	None	coupon_id → coupons.coupon_id	[PURPOSE]: Marketing promotional coupon codes, percentage discount values, minimum order thresholds, and expiration dates. [DOMAIN]: Domain M: Loyalty, Customer Retention & Rewards. [GRAIN]: One row per discount coupon definition (coupon_id). [TIER & REFRESH]: GOLD_CURATED

Domain O: Product Information Management (PIM) & Merchandising (8 Tables)

Table Name	Record Count	Primary Key	Foreign Key Links	Business Description
product_attribute_definitions	4	None	attribute_id → attributes.attribute_id	[PURPOSE]: Master Product Information Management (PIM) attribute definitions (e.g. skin_type, battery_capacity, fabric_composition). [DOMAIN]: Domain O: Product Information Management (PIM). [GRAIN]: One row per attribute definition (attribute_id). [TIER & REFRESH]: GOLD_CURATED
product_attribute_values	2,000	None	attribute_id → attributes.attribute_id product_id → products.product_id	[PURPOSE]: EAV (Entity-Attribute-Value) structured technical product specifications and filter criteria for catalog search. [DOMAIN]: Domain O: Product Information Management (PIM). [GRAIN]: One row per SKU attribute value assignment (value_id). [TIER & REFRESH]: SILVER_CONSOLIDATED
product_media_gallery	1,500	None	media_id → medias.media_id product_id → products.product_id	[PURPOSE]: High-resolution product images, video URLs, CDN asset paths, and display sort orders. [DOMAIN]: Domain O: Product Information Management (PIM). [GRAIN]: One row per media visual asset (media_id). [TIER & REFRESH]: SILVER_CONSOLIDATED
product_multilingual_translations	1,500	None	product_id → products.product_id translation_id → translations.translation_id	[PURPOSE]: Localized product titles and marketing descriptions translated into French, German, Italian, Spanish, and Dutch. [DOMAIN]: Domain O: Product Information Management (PIM). [GRAIN]: One row per SKU per language translation (translation_id). [TIER & REFRESH]: GOLD_CURATED
product_size_charts	3	None	category_id → categories.category_id size_chart_id → size_charts.size_chart_id	[PURPOSE]: Category size conversion tables, body measurements in centimeters, and international size mapping. [DOMAIN]: Domain O: Product Information Management (PIM). [GRAIN]: One row per size chart specification (size_chart_id). [TIER & REFRESH]: SILVER_CONSOLIDATED
product_brand_guidelines	3	None	guideline_id → guidelines.guideline_id	[PURPOSE]: Manufacturer minimum advertised price (MAP) policies, brand authorization rules, and trademark guidelines.

Table Name	Record Count	Primary Key	Foreign Key Links	Business Description
				[DOMAIN]: Domain O: Product Information Management (PIM). [GRAIN]: One row per brand guideline (guideline_id). [TIER & REFRESH]: SILVER_CONSOLIDATED
category_hierarchy_paths	4	None	category_id → categories.category_id parent_category_id → parent_categorys.parent_category_id	[PURPOSE]: Materialized category hierarchy breadcrumb paths for fast e-commerce storefront navigation and facet filtering. [DOMAIN]: Domain O: Product Information Management (PIM). [GRAIN]: One row per hierarchical category path (path_id). [TIER & REFRESH]: GOLD_CURATED
seo_meta_tags_registry	2	None	seo_id → seos.seo_id	[PURPOSE]: Search engine optimization (SEO) title tags, meta descriptions, and canonical URLs across all web catalog pages. [DOMAIN]: Domain O: Product Information Management (PIM). [GRAIN]: One row per indexed page URL (seo_id). [TIER & REFRESH]: SILVER_CONSOLIDATED

Domain P: Retail Physical Stores & Omni-Channel POS (8 Tables)

Table Name	Record Count	Primary Key	Foreign Key Links	Business Description
physical_store_locations	4	None	store_id → stores.store_id	[PURPOSE]: Master directory of brick-and-mortar retail stores, address geolocations, floor square meters, and opening status. [DOMAIN]: Domain P: Retail Physical Stores & Omni-Channel POS. [GRAIN]: One row per retail store branch (store_id). [TIER & REFRESH]: GOLD_CURATED
pos_terminal_registers	5	None	register_id → registers.register_id store_id → stores.store_id	[PURPOSE]: Point of Sale (POS) checkout terminal registers, hardware models, serial numbers, and IP address bindings. [DOMAIN]: Domain P: Retail Physical Stores & Omni-Channel POS. [GRAIN]: One row per POS terminal register (register_id). [TIER & REFRESH]: SILVER_CONSOLIDATED
pos_store_transactions	1,500	None	cashier_employee_id → cashier_employees.cashier_employee_id pos_transaction_id → pos_transactions.pos_transaction_id	[PURPOSE]: In-store brick-and-mortar point-of-sale customer sales transactions, cashier employee IDs, and payment tender types. [DOMAIN]: Domain P: Retail Physical Stores & Omni-Channel POS. [GRAIN]: One row per in-store checkout transaction header (pos_transaction_id). [TIER & REFRESH]: GOLD_CURATED
pos_transaction_items	2,500	None	pos_item_id → pos_items.pos_item_id pos_transaction_id → pos_transactions.pos_transaction_id	[PURPOSE]: Granular purchase line items sold through physical store cash registers. [DOMAIN]: Domain P: Retail Physical Stores & Omni-Channel POS. [GRAIN]: One row per in-store purchased item line (pos_item_id). [TIER & REFRESH]: SILVER_CONSOLIDATED
store_inventory_levels	2,000	None	product_id → products.product_id store_id → stores.store_id	[PURPOSE]: Real-time store on-hand stock quantities, shelf availability, and periodic cycle count audit dates per retail location. [DOMAIN]: Domain P: Retail Physical Stores & Omni-Channel POS. [GRAIN]: One row per SKU per retail store (store_stock_id). [TIER & REFRESH]: GOLD_CURATED
store_employee_rosters	200	None	roster_id → rosters.roster_id store_id → stores.store_id	[PURPOSE]: In-store retail employee staffing rosters, shift schedules, and operational store roles (Cashier, Floor Supervisor). [DOMAIN]: Domain P: Retail Physical Stores & Omni-Channel POS. [GRAIN]: One row per store employee shift (roster_id). [TIER & REFRESH]: SILVER_CONSOLIDATED
store_cash_drawer_counts	100	None	drawer_count_id → drawer_counts.drawer_count_id register_id → registers.register_id	[PURPOSE]: Daily store register cash drawer end-of-shift balancing logs and physical cash counting variances in EUR. [DOMAIN]: Domain P: Retail Physical Stores & Omni-Channel POS. [GRAIN]: One row per register drawer cash count (drawer_count_id). [TIER & REFRESH]: SILVER_CONSOLIDATED
click_and_collect_orders	80	None	bopis_id → bopiss.bopis_id order_id → orders.order_id	[PURPOSE]: Buy Online Pick Up in Store (BOPIS) orders, store pickup verification PINs, customer collection

Table Name	Record Count	Primary Key	Foreign Key Links	Business Description
				timestamps, and fulfillment states. [DOMAIN]: Domain P: Retail Physical Stores & Omni-Channel POS. [GRAIN]: One row per BOPIS pickup order (bopis_id). [TIER & REFRESH]: GOLD_CURATED

 Domain Q: Non-Production, Sandboxes & QA Archives (11 Tables)

Table Name	Record Count	Primary Key	Foreign Key Links	Business Description
dev_customer_churn_feature_store	1,500	None	user_id → users.user_id	[PURPOSE]: Machine learning feature store containing customer behavioral aggregations and predicted churn risk scores. [DOMAIN]: Domain Q: Non-Production, Sandbox & QA Archives. [GRAIN]: One row per customer feature vector (user_id). [TIER & REFRESH]: SANDBOX_QA
dev_product_embedding_vectors	150	None	product_id → products.product_id	[PURPOSE]: Product catalog semantic embedding vector embeddings for experimental visual similarity models. [DOMAIN]: Domain Q: Non-Production, Sandbox & QA Archives. [GRAIN]: One row per product vector embedding (product_id). [TIER & REFRESH]: SANDBOX_QA
sandbox_dynamic_pricing_sim_v1	80	None	product_id → products.product_id sim_id → sims.sim_id	[PURPOSE]: Offline dynamic pricing simulator sandbox estimating price elasticity curves and revenue trade-offs. [DOMAIN]: Domain Q: Non-Production, Sandbox & QA Archives. [GRAIN]: One row per pricing simulation run (sim_id). [TIER & REFRESH]: SANDBOX_QA
sandbox_search_ranking_ab_test	60	None	experiment_id → experiments.experiment_id	[PURPOSE]: Search ranking algorithm A/B experiment evaluation logs recording NDCG@10 relevance scores. [DOMAIN]: Domain Q: Non-Production, Sandbox & QA Archives. [GRAIN]: One row per search experiment evaluation (experiment_id). [TIER & REFRESH]: SANDBOX_QA
qa_checkout_synthetic_fuzz_tests	300	None	fuzz_id → fuzzs.fuzz_id	[PURPOSE]: Automated checkout synthetic fuzz testing payloads and HTTP response code validations. [DOMAIN]: Domain Q: Non-Production, Sandbox & QA Archives. [GRAIN]: One row per fuzz test execution (fuzz_id). [TIER & REFRESH]: SANDBOX_QA
qa_load_test_sessions_backup	1,200	None	test_run_id → test_runs.test_run_id virtual_user_id → virtual_users.virtual_user_id	[PURPOSE]: Archived load test benchmark traffic logs generated by synthetic load testing tools. [DOMAIN]: Domain Q: Non-Production, Sandbox & QA Archives. [GRAIN]: One row per load test synthetic run (test_run_id). [TIER & REFRESH]: SANDBOX_QA
legacy_orders_2023_archive	1,500	None	legacy_order_id → legacy_orders.legacy_order_id	[PURPOSE]: Deprecated archive of historical 2023 customer order headers retained for legal and compliance audit. [DOMAIN]: Domain Q: Non-Production,

Table Name	Record Count	Primary Key	Foreign Key Links	Business Description
				Sandbox & QA Archives. [GRAIN]: One row per archived legacy order (legacy_order_id). [TIER & REFRESH]: SANDBOX_QA
legacy_products_deprecated	300	None	legacy_product_id → legacy_products.legacy_product_id	[PURPOSE]: Discontinued catalog products and end-of-life merchandise from previous retail seasons. [DOMAIN]: Domain Q: Non-Production, Sandbox & QA Archives. [GRAIN]: One row per discontinued SKU (legacy_product_id). [TIER & REFRESH]: SANDBOX_QA
test_fraud_mock_transactions	300	None	mock_id → mocks.mock_id	[PURPOSE]: Synthetic fraud model test transactions evaluating rule triggers and mock transaction risks. [DOMAIN]: Domain Q: Non-Production, Sandbox & QA Archives. [GRAIN]: One row per mock transaction (mock_id). [TIER & REFRESH]: SANDBOX_QA
test_carrier_webhook_payloads	300	None	test_payload_id → test_payloads.test_payload_id	[PURPOSE]: Synthetic logistics carrier webhook testing fixtures for development mocking and contract verification. [DOMAIN]: Domain Q: Non-Production, Sandbox & QA Archives. [GRAIN]: One row per synthetic webhook payload (test_payload_id). [TIER & REFRESH]: SANDBOX_QA
agent_worklog_shifts	75	None	agent_id → agents.agent_id shift_id → shifts.shift_id	[PURPOSE]: Daily customer support agent productivity records, tickets resolved, and average handle time (AHT) in seconds. [DOMAIN]: Domain J: Customer Support, CRM & CSAT. [GRAIN]: One row per agent shift (shift_id). [TIER & REFRESH]: SILVER_CONSOLIDATED